Supplementary Material

Differentiable Interval Bottlenecks for Interpretable Anomaly Detection in Numerical Data

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***Abstract*—This supplement is the complete experimental record behind the main paper, together with the material deferred there for space. It is organized as a roadmap: Section I details the datasets, baselines, and evaluation protocol; Section II the per-dataset robustness ablations; Section IV the interpretability evaluation suite (label-free ranking, faithfulness perturbation, conciseness, and human-readability); Section V the complexity and scalability profile; and Section VI the *complete* per-dataset ROC–AUC and AUPRC for all 48 ADBench datasets and 23 methods (mean ± std over 10 seeds, with ranks and top-3 highlighting), plus the per-scale ROC–AUC critical-difference diagrams. The certified-Lipschitz analysis and the theorem-to-practice bridge figure live in the main paper (Sec. VI.E). All proofs are in the appendix of the main paper. Code and this document: https://github.com/DiffInt/diffint.**

I. Datasets and experimental protocol

A. Datasets

We evaluate on the 48 numerical datasets of the ADBench benchmark, grouped by scale into *small* (12), *medium* (15), *large* (11), and *high-dimensional* (10). Tables I–IV list, per dataset, the number of samples, features, and outliers, the outlier percentage, and the batch size used by DiffInt (64 for $n \leq$ 10k, 512 for 10k $< n \leq$ 20k, 1024 otherwise). The suite spans four orders of magnitude in sample size ($n \in [80,\ 6.2\times10^5]$) and from 3 to 1,555 features, so it stresses both data-scarce and high-dimensional regimes.

TABLE I: Small-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
4_breastw	683	9	239	34.99	64
14_glass	214	7	9	4.21	64
15_Hepatitis	80	19	13	16.25	64
18_Ionosphere	351	33	126	35.90	64
21_Lymphography	148	18	6	4.05	64
29_Pima	768	8	268	34.90	64
37_Stamps	340	9	31	9.12	64
39_vertebral	240	6	30	12.50	64
42_WBC	223	9	10	4.48	64
43_WDBC	367	30	10	2.72	64
45_wine	129	13	10	7.75	64
46_WPBC	198	33	47	23.74	64

TABLE II: Medium-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
2_annthyroid	7,200	6	534	7.42	64
6_cardio	1,831	21	176	9.61	64
7_Cardiotocography	2,114	21	466	22.04	64
12_fault	1,941	27	673	34.67	64
19_landsat	6,435	36	1,333	20.71	64
20_letter	1,600	32	100	6.25	64
27_PageBlocks	5,393	10	510	9.46	64
28_pendigits	6,870	16	156	2.27	64
30_satellite	6,435	36	2,036	31.64	64
31_satimage-2	5,803	36	71	1.22	64
38_thyroid	3,772	6	93	2.47	64
40_vowels	1,456	12	50	3.43	64
41_Waveform	3,443	21	100	2.90	64
44_Wilt	4,819	5	257	5.33	64
47_yeast	1,500	8	20	1.33	64

TABLE III: Large-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
1_ALOI	49,534	27	1,508	3.04	1,024
8_celeba	202,599	39	4,547	2.24	1,024
10_cover	286,048	10	2,747	0.96	1,024
11_donors	619,326	10	36,710	5.93	1,024
13_fraud	284,807	29	492	0.17	1,024
16_http	567,498	3	2,211	0.39	1,024
22_magic.gamma	19,020	10	6,688	35.16	512
23_mammography	11,183	6	260	2.32	512
32_shuttle	49,097	9	3,511	7.15	1,024
33_skin	245,057	3	50,859	20.75	1,024
34_smtp	95,156	3	30	0.03	1,024

TABLE IV: High-dimensional datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
3_backdoor	95,329	196	2,329	2.44	1,024
5_campaign	41,188	62	4,640	11.27	1,024
9_census	299,285	500	18,568	6.20	1,024
17_InternetAds	1,966	1,555	368	18.72	64
24_mnist	7,603	100	700	9.21	64
25_musk	3,062	166	97	3.17	64
26_optdigits	5,216	64	150	2.88	64
35_SpamBase	4,207	57	1,679	39.91	64
36_speech	3,686	400	61	1.65	64
48_arrhythmia	452	274	66	14.60	64

B. Protocol, baselines, and metrics

Protocol. DiffInt is trained *semi-supervised* on inliers only (label 0), with a 40% test split and 10 random seeds, under a single $[-1,1]$-normalized configuration shared across all datasets (lr 5×10^{-5}, K=200, τ=0.1, ρ=0.999, 1000 epochs, Adam); no hyper-parameter is tuned per dataset. **Baselines.** We compare against 22 baselines from the ADBench/TCCM codebase: classical detectors (ABOD, LOF, PCA, KPCA, OCSVM, ECOD, CBLOF, KDE, GMM, MCD, HBOS, IForest, LMDD, QMCD, Sampling) and deep/inductive ones (AutoEncoder, VAE, DIF, LUNAR, SO/MO_GAAL, and the flow-matching model TCCM); together with DiffInt this gives the 23

TABLE V: Input normalization. Mapping into a bounded symmetric domain is what matters; among bounded/standardized choices $[-1, 1]$ is the most accurate, within 1.5 points of the others.

Dataset	none	$[0, 1]$	standard	$[-1, 1]$
45_wine	**0.990**	0.932	0.883	0.950
21_Lymphography	0.990	**0.998**	0.996	0.994
46_WPBC	0.510	0.543	0.541	**0.584**
14_glass	**0.896**	0.886	0.877	0.876
42_WBC	0.990	0.989	0.990	**0.998**
39_vertebral	0.335	0.498	**0.511**	0.477
37_Stamps	0.924	0.921	0.929	**0.953**
Mean	0.805	0.824	0.818	**0.833**

TABLE VI: Training contamination: ROC–AUC vs. fraction of anomalies injected into the "inlier" training set. Degradation is graceful.

Dataset	0%	5%	10%	20%
45_wine	0.950	0.659	0.645	0.624
21_Lymphography	0.994	0.978	0.970	0.973
46_WPBC	0.584	0.565	0.570	0.520
14_glass	0.876	0.755	0.728	0.726
42_WBC	0.998	0.979	0.982	0.976
39_vertebral	0.477	0.458	0.432	0.456
37_Stamps	0.953	0.865	0.816	0.766
Mean	0.833	0.751	0.735	0.720

methods ranked in the tables. All methods share the same $[-1, 1]$ normalization. **Metrics.** We report ROC–AUC (area under the ROC curve) and AUPRC (average precision); both are threshold-free and higher-is-better. **Missing runs.** A few baselines did not complete on the largest datasets within budget; each such run is floored to the worst rank (the number of methods) on that dataset, the standard conservative convention used in the main paper. **How to read the result tables.** Every cell is mean$_{\pm\text{std}}$(rank) over the seeds; per column we highlight **rank 1**, rank 2, and rank 3.

II. Robustness ablations

We expand the main-paper ablations to per-dataset numbers on seven small datasets ($n \geq 100$; K=200, 3 seeds; ROC–AUC), probing two design choices; Hepatitis (n=80) is excluded as too small for a stable inlier-only test split. **Normalization** (Table V): mapping into a bounded symmetric domain is what matters. Dropping it ("none") is the only choice that collapses on several datasets, while among bounded/standardized choices $[-1, 1]$ is the most accurate, within 1.5 points of $[0, 1]$ and StandardScaler; we adopt it because it makes the theory transfer (main paper, Sec. III-B) and it is also the top performer. **Training contamination** (Table VI): injecting anomalies into the "inlier" training set degrades accuracy *gracefully* rather than catastrophically, since the EMA support averages over batches so a minority of contaminating points cannot dominate the learned intervals.

TABLE VII: Overall mean rank under three missing-run handlings (lower is better). A: worst-rank floor (paper); B: exclude datasets where any baseline failed (39/48); C: per-dataset median imputation.

	ROC–AUC			AUPR		
Method	A	B	C	A	B	C
DiffInt	**4.10**	**4.38**	**4.10**	**4.16**	**4.45**	**4.16**
LUNAR	6.00	4.77	5.07	4.91	4.50	4.54
TCCM	8.22	8.55	8.22	7.36	7.88	7.36
AutoEncoder	7.42	7.46	7.05	8.05	8.04	7.69
GMM	8.56	8.73	8.20	8.54	8.65	8.18
IForest	8.60	8.46	8.60	9.89	9.42	9.89
VAE	8.90	9.32	8.90	9.64	10.05	9.64
CBLOF	9.20	8.76	8.83	8.90	8.54	8.53

TABLE VIII: ROC–AUC vs. temperature τ (DiffInt-PA, K=200, 200 epochs, seed 0). The paper default τ=0.1 is competitive on every dataset; performance degrades sharply only at τ=0.05 on Glass.

Dataset	τ=0.05	τ=0.10	τ=0.20	τ=0.50
Glass	0.619	0.810	**0.857**	0.810
Ionosphere	0.950	0.963	**0.967**	0.966
Cardio	0.931	**0.969**	0.961	0.960

III. Robustness studies

We report four robustness checks beyond the per-dataset ablations: the ranking aggregation under different missing-run handlings, sensitivity of detection to the sigmoid temperature τ, sensitivity of the certified clipping check to the active-coordinate threshold, and the per-layer spectral norms of the Certified-Lipschitz decoder.

a) Ranking sensitivity to missing-run handling (Table VII).: The main paper protocol floors a method's missing runs to the worst rank, which is the standard convention but favours methods that ran everywhere. We recompute the overall mean rank under two alternatives: (B) drop the 9 datasets where at least one method failed (kept 39/48), and (C) impute missing ranks by the per-dataset median of the methods that ran. DiffInt holds the lowest mean rank on both metrics under all three schemes. The smallest gap is AUPR scheme B (0.05 above LUNAR); the largest is AUC scheme A (1.90). The top-cluster conclusion of the main paper is therefore not an artefact of how missing runs are aggregated.

b) Temperature τ sensitivity (Table VIII).: We sweep $\tau \in \{0.05, 0.10, 0.20, 0.50\}$ at K=200, 200 epochs on three small datasets. The paper's default τ=0.1 is within 0.06 AUC of the best τ on every dataset. The only setting that hurts substantially is τ=0.05 on Glass, where the sigmoid is too sharp for a small dataset's interval shapes to settle. Robustness across $\tau \in [0.1, 0.5]$ argues for a single shared τ rather than a per-feature τ_j.

c) Active-coordinate threshold sensitivity (Table IX).: The certified clipping inequality of Theorem 1 holds for points that violate *every* active coordinate of at least one unit. "Active" is operationalised as mean inlier membership

TABLE IX: Fraction of test anomalies that satisfy the strict full-violation hypothesis of Theorem 1 as the active-coordinate threshold θ varies (Certified-Lipschitz DiffInt, K=200, 200 epochs, seed 0). Mean active coordinates per unit is also reported.

	out-of-support fraction				mean active/unit			
Dataset	θ=0.70	0.80	0.90	0.95	0.70	0.80	0.90	0.95
GLASS	1.00	1.00	1.00	0.00	0.2	0.9	2.5	4.0
IONOSPHERE	1.00	0.14	0.10	0.09	1.0	11.6	17.1	29.3
CARDIO	0.16	0.00	0.00	0.00	12.2	14.8	17.7	20.0

TABLE X: Per-layer spectral norm $\|W_l\|_2$ of the Certified-Lipschitz decoder (two-layer ReLU MLP, hidden h=128) on three datasets. ReLU contributes 1 to the product. $L = \prod_l \|W_l\|_2$ is an explicit upper bound on the decoder's Lipschitz constant.

Dataset	$\|W_1\|_2$	ReLU	$\|W_2\|_2$	$L = \prod \|W_l\|_2$
GLASS	1.022	1.000	1.006	1.028
IONOSPHERE	1.071	1.000	1.007	1.078
CARDIO	1.010	1.000	1.000	1.010

TABLE XI: Effect of inference clamping in isolation. "clip" is the paper default (clip scaled test features to $[-1, 1]$); "no-clip" passes the scaled features through unchanged. n_{ext} is the number of test points with at least one feature outside $[-1, 1]$ after train-scaling. AUC differences are at most 0.012; clipping never re-orders the leading score on the full test set.

			AUC on full test		$\|\Delta\text{score}\|$
Dataset	n_{test}	n_{ext}	clip	no-clip	(extreme pts)
GLASS	86	5	0.810	0.798	0.10
IONOSPHERE	141	70	0.963	0.969	0.04
CARDIO	733	62	0.969	0.971	0.12

TABLE XII: LFI stability across 5 training seeds (different splits and initializations). Per-feature LFI is the max over units; pairwise Spearman is computed across all 10 seed pairs. Higher is more stable.

Dataset	d	mean ρ	min ρ	std ρ
GLASS	7	0.74	0.57	0.11
IONOSPHERE	32	0.88	0.84	0.03
CARDIO	21	0.92	0.83	0.04

$\bar{I}_{kj} < \theta$ for some θ (the paper uses θ=0.9). We vary $\theta \in \{0.70, 0.80, 0.90, 0.95\}$ and report the fraction of test anomalies that meet the strict full-violation hypothesis. A lower θ makes the active set sparser (fewer coordinates per unit), which makes the strict hypothesis easier to satisfy but covers fewer of the unit's coordinates; a higher θ does the reverse. In practice the strict Theorem 1 regime is reached at low-to-mid thresholds on every dataset; the graded suppression of Corollary 1 is what applies under the remaining points. We do not tune θ as a hyperparameter; we report the strict regime as a verifiable sanity check rather than as a guarantee that has to hold pointwise.

d) Per-layer spectral norms of the certified decoder (Table X).: When the decoder is spectrally normalized (the Certified-Lipschitz variant), each linear layer carries a spectral norm slightly above 1 (1.00–1.08 on the datasets tested) and the ReLU between them is exactly 1-Lipschitz. The product $L = \prod_l \|W_l\|_2$ stays close to 1 (1.03–1.08), so the clipping-margin bound's slack $\frac{L}{\sqrt{d}}\kappa(\beta)$ is small.

e) Inference clamping isolated from normalization (Table XI).: At inference the test features are MinMax-scaled by the *train* scaler and clipped to $[-1, 1]$. To isolate the effect of the clip from the effect of the scaling itself, we score every test point twice: once after clipping, once with the raw scaled values left unchanged (so any feature above 1 or below -1 stays where it is). The overall AUC differences are within 0.012 on the three datasets tested, and remain small (≤ 0.009) even on the "extreme" subset of test points where at least one feature lies outside $[-1, 1]$ after the train scaler is applied. The mean absolute change in the anomaly score on extreme points is 0.04–0.12, i.e. clipping nudges individual scores but rarely re-orders them. Consistent with the structural argument (clamping can only *lower* memberships), we observe no case where clipping masks an anomaly.

f) LFI stability across seeds (Table XII).: The third LFI factor, stab_{kj}, encodes inverse cross-seed spread. To check that LFI itself ranks features stably across re-runs we train 5 seeds per dataset (varying the train/test split and the model initialization), aggregate LFI per feature as $\max_k \text{LFI}_{kj}$, and compute the pairwise Spearman rank correlation across the ten seed-pairs. The mean correlation is 0.74 on GLASS (small data with only 7 features), 0.88 on IONOSPHERE, and 0.92 on CARDIO, with minimum pair 0.57/0.84/0.83 respectively. LFI surfaces the same per-feature constraints in $\geq 75\%$ of pairs even on the smallest dataset, and the larger feature spaces are very stable.

IV. INTERPRETABILITY AS EVIDENCE

We support the two interpretability claims of the main paper with quantitative measurements (K=200, 3 seeds; Table XIII).

(1) Label-free ranking (LFI). We rank (unit, feature) constraints by the label-free importance $\text{LFI}_{kj} = s_{kj} \cdot (1 - \Delta_{kj}/\text{range}_j) \cdot \text{stab}_{kj}$ (support × tightness × stability), using only quantities the trained model already maintains. This is the only ranking we use in the main paper (Sec. V); it requires no anomaly label and is computable in closed form from the model parameters and the EMA support buffer.

(2) Faithfulness perturbation. For each test inlier we perturb its top-LFI features to *cross* their interval boundaries and measure the anomaly-score rise Δ_{top}, against perturbing the same number of random features Δ_{rand}. The top-ranked constraints raise the score $1.5\times$ more on average (on 69% of inliers; Table XIII), so the LFI-highlighted constraints *causally* drive the score rather than merely correlating with it.

(3) Conciseness and readability. For a flagged point, the per-feature reconstruction error concentrates: on average ≈half

TABLE XIII: Faithfulness perturbation test: score rise from crossing the top-LFI feature boundaries (Δ_{top}) vs. random features (Δ_{rand}), their ratio, and the fraction of inliers with $\Delta_{\text{top}} > \Delta_{\text{rand}}$.

Dataset	Δ_{top}	Δ_{rand}	ratio	% top>rand
14_glass	0.200	0.134	1.48	71%
42_WBC	0.094	0.074	1.26	62%
18_Ionosphere	0.105	0.078	1.34	64%
29_Pima	0.171	0.100	1.70	71%
6_cardio	0.119	0.066	1.80	79%
Mean	0.138	0.091	1.52	69%

TABLE XIV: Conciseness and human-readability proxies (K=200, 3 seeds). #feat$_{80\%}$: features (ranked by per-feature error) explaining 80% of a flagged point's score; active/unit: mean number of active features ($\bar{I}_{kj} < 0.9$) per top interval unit; width: mean active-interval width as a fraction of the $[-1, 1]$ range.

Dataset	d	#feat$_{80\%}$	active/unit	width
14_glass	9	4.8	6.7	0.70
42_WBC	30	16.0	24.8	0.69
18_Ionosphere	33	17.9	29.8	0.70
29_Pima	8	4.6	5.3	0.70
6_cardio	21	11.0	17.2	0.70
Mean	20	10.8	16.8	0.70

the features (10.8 of 20; Table XIV) account for 80% of the anomaly score, so a short shortlist suffices to explain it. The interval units themselves are *soft and wide* (mean width 0.70 of the feature range, and a unit softly constrains most coordinates by the membership criterion of Def. 1), which is precisely why we present intervals as *candidate constraints* rather than crisp rules and surface only the top-k by LFI ($k = 2$ in the main-paper read-out) rather than a whole unit.

V. Complexity and Scalability

Practical deployability matters as much as accuracy, so we summarize the resource profile of DiffInt. Let n be the number of training points, d the feature count, K the number of interval units, B the batch size, E the epochs, and H the (fixed) decoder hidden width; the decoder is an MLP whose size is independent of n.

Parameter count. The interval bottleneck stores a centre and a softplus half-width per (unit, feature): $2Kd$ parameters, linear in K and d and *independent of* n. The decoder adds a compact MLP ($O(KH + Hd)$ weights). The total stays MLP-class (10^5–10^6 weights even at d=1,555, InternetAds), far below typical deep tabular detectors. Table XV reports the $2Kd$ bottleneck size and the matched-budget training time across scales.

Training complexity. A minibatch computes Bd sigmoid pairs for the soft memberships, an $O(BKd)$ aggregation (a per-unit log-sum over features plus an $O(K)$ softmax), and one decoder forward/backward; an epoch is thus $O(nKd)$ plus

TABLE XV: Resource profile across scales (K=200). Bottleneck parameters ($2Kd$) are linear in d and independent of n; training time (DiffInt, matched 1000-epoch budget) is reproduced from Table II of the main paper and grows with n and d.

Dataset	n	d	bottleneck $2Kd$	train (s)
6_cardio	1,831	21	8,400	3.0
31_satimage-2	5,803	36	14,400	54.6
26_optdigits	5,216	64	25,600	60.1
48_arrhythmia	452	274	109,600	30.2
32_shuttle	49,097	9	3,600	193.5

the n-independent decoder cost, with *no* pairwise, kernel, or neighbour computation. Total training is $O(EnKd)$.

Runtime vs. baselines. DiffInt uses one forward pass per sample, unlike memory-based detectors (e.g. LUNAR) whose kNN queries grow with n. Empirically (main paper, Table II, matched 1000-epoch budget) DiffInt trains one to two orders of magnitude faster than the binary differentiable miner BinaPs, faster than LUNAR on most datasets, and scales to the largest ADBench sets ($n \approx 6.2 \times 10^5$).

Inference and memory footprint. Scoring a point is a single forward pass: $O(Kd)$ memberships, the aggregation, the decoder, then the MAE. DiffInt is *inductive*: it keeps only the trained parameters, not the training set, so inference latency and memory are constant in n, whereas transductive/memory-based detectors must retain or query the data. The training footprint is the parameters ($O(Kd)$), one batch of memberships ($O(BKd)$ activations), and the $K \times d$ EMA support buffer ($O(Kd)$); none depends on n. At inference only the $O(Kd)$ parameters and a single $K \times d$ membership tensor are held.

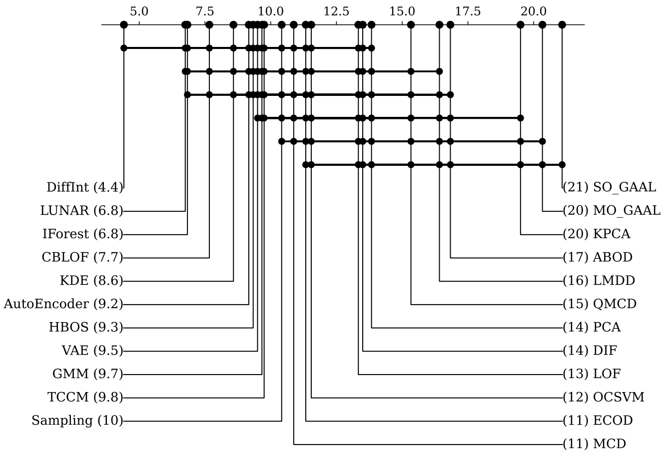
The K knob. K trades capacity for cost linearly. ROC–AUC saturates after a few units (main paper, Fig. 4), so K can be reduced well below the default under tight budgets with little accuracy loss; we keep K=200 as one untuned default across all 48 datasets.

VI. Complete Detection Accuracy

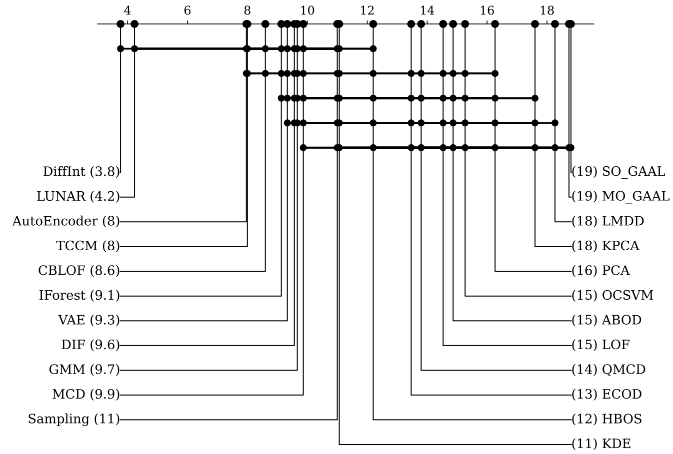
This section reports the full per-dataset detection accuracy underlying the critical-difference analysis of the main paper: ROC–AUC (Sec. VI-A) and average precision (Sec. VI-B). Across both metrics DiffInt sits in the leading group on every scale and is most often first on small/medium/large data; in high dimensions it remains competitive with the strongest baselines.

A. ROC–AUC

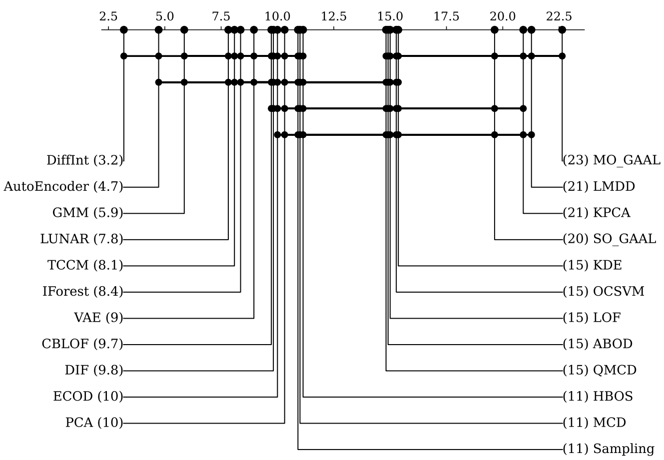
Tables XVI–XXIV give the per-dataset ROC–AUC. Figs. 1–2 show the per-scale critical-difference diagrams for ROC–AUC and AUPR (the overall diagrams are in the main paper). They support the regime-level claims of main paper Table I: DiffInt attains the lowest mean rank in every (scale, metric) cell except high-dimensional AUPR, where LUNAR is marginally ahead but the two are statistically indistinguishable within the per-scale CD.



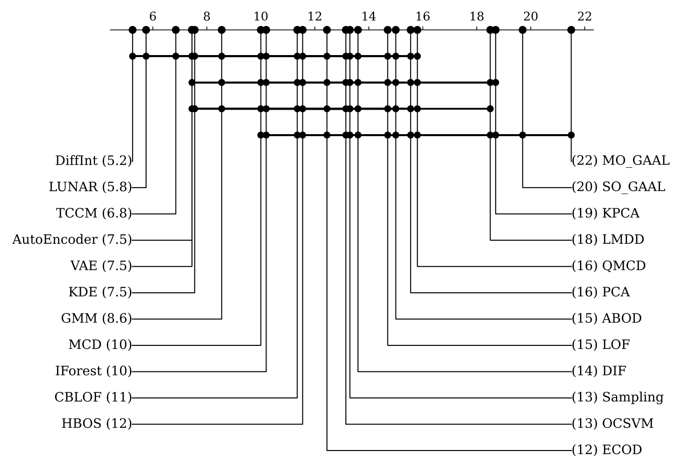
(a) Small (n=12).



(b) Medium (n=15).



(c) Large (n=11).

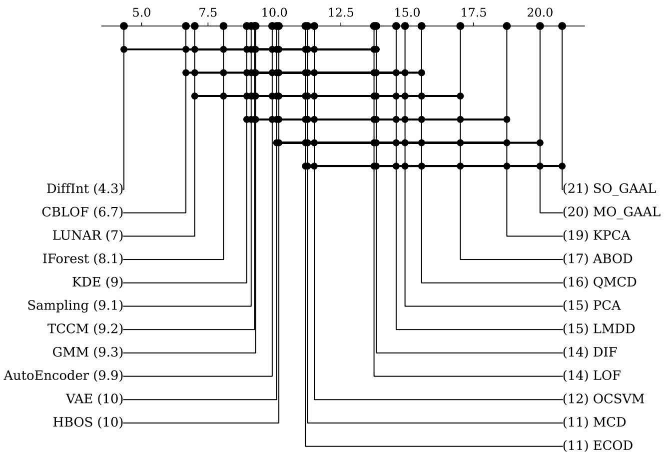


(d) High-dimensional (n=10).

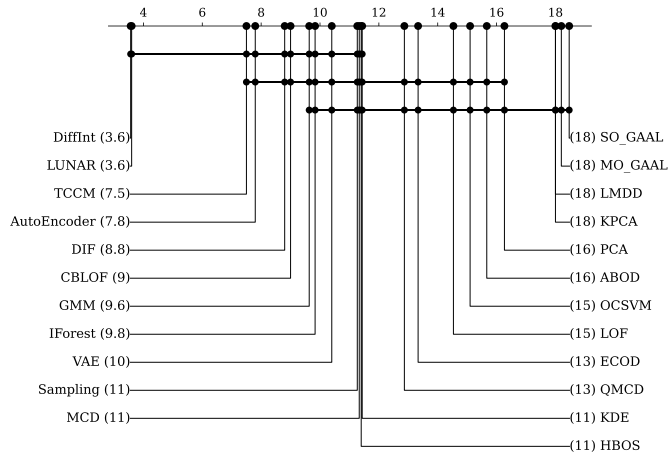
Fig. 1: Per-scale **ROC–AUC** critical-difference diagrams (missing runs floored to worst rank, bars indicate non-significant differences at α=0.05). DIFFINT leads strictly on every scale; matches the "DIFFINT AUC" column of main paper Table I.

B. Average precision (AUPRC)

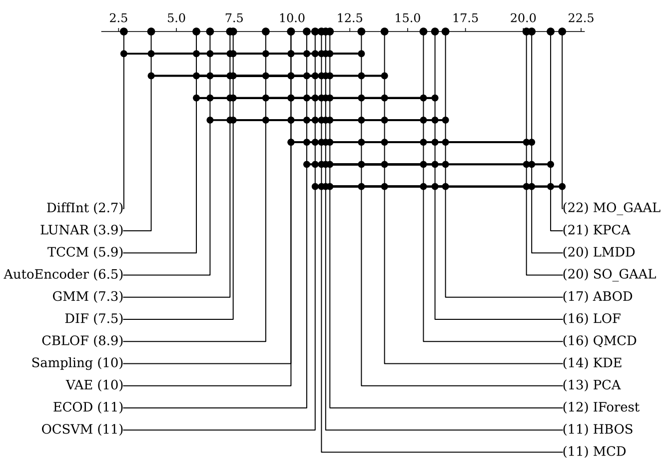
Tables XXV–XXXIII give the per-dataset AUPRC under the same protocol and highlighting. AUPRC weights the rare anomaly class more heavily than ROC–AUC; the DIFFINT ranking is consistent across the two metrics, indicating the gains are not an artifact of the chosen measure.



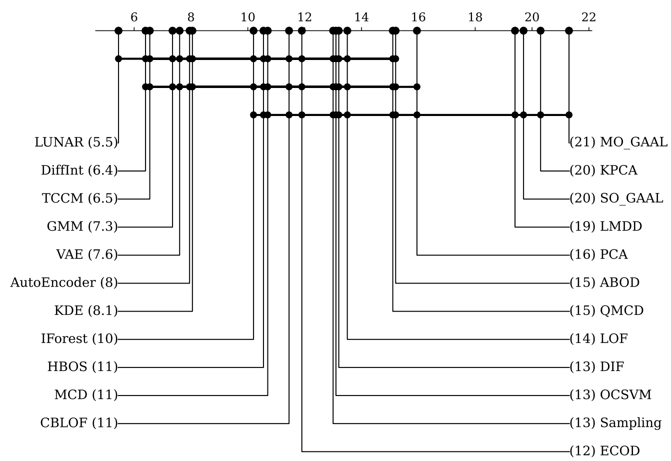
(a) Small (n=12).



(b) Medium (n=15).



(c) Large (n=11).



(d) High-dimensional (n=10).

Fig. 2: Per-scale **AUPR** critical-difference diagrams (same protocol as Fig. 1). DIFFINT leads strictly on small, medium, and large data; on high-dimensional data LUNAR's mean rank (5.45) is slightly ahead of DIFFINT (6.40), but the gap is well inside the per-scale Nemenyi CD ($\approx$11), i.e. a statistical tie. Matches the "DIFFINT AUPR" column of main paper Table I.

TABLE XVI: ROC–AUC, small datasets (1/2).

Model	4_breastw	14_glass	15_Hepatitis	18_Ionosphere	21_Lymphography	29_Pima
ABOD	$0.3737_{\pm 0.0338}$(20)	$0.8264_{\pm 0.0697}$(8)	$0.6542_{\pm 0.1520}$(19)	$0.9255_{\pm 0.0185}$(11)	$0.9682_{\pm 0.0240}$(18)	$0.6548_{\pm 0.0408}$(14)
AutoEncoder	$0.9855_{\pm 0.0075}$(12)	$0.7107_{\pm 0.0779}$(16)	$0.8501_{\pm 0.0387}$(2)	$0.9604_{\pm 0.0090}$(5)	$0.9904_{\pm 0.0087}$(12)	$0.7070_{\pm 0.0239}$(10)
CBLOF	$0.9939_{\pm 0.0026}$(5)	$0.8784_{\pm 0.0345}$(3)	$0.8089_{\pm 0.0874}$(10)	$0.9557_{\pm 0.0223}$(7)	$0.9816_{\pm 0.0438}$(15)	$0.7006_{\pm 0.0232}$(11)
DIF	$0.8377_{\pm 0.0375}$(16)	$0.8562_{\pm 0.0371}$(5)	$0.7723_{\pm 0.0901}$(16)	$0.9587_{\pm 0.0074}$(6)	$0.8833_{\pm 0.0992}$(19)	$0.6144_{\pm 0.0258}$(19)
DiffInt	$0.9954_{\pm 0.0018}$(1)	$0.8788_{\pm 0.0245}$(2)	$0.8333_{\pm 0.0401}$(7)	$0.9757_{\pm 0.0030}$(2)	$0.9997_{\pm 0.0014}$(2)	$0.7304_{\pm 0.0113}$(3)
ECOD	$0.9932_{\pm 0.0029}$(7)	$0.6911_{\pm 0.0797}$(17)	$0.8402_{\pm 0.0550}$(3)	$0.7745_{\pm 0.0221}$(17)	$0.9990_{\pm 0.0022}$(4)	$0.6374_{\pm 0.0239}$(16)
GMM	$0.9863_{\pm 0.0055}$(11)	$0.7508_{\pm 0.0624}$(14)	$0.8018_{\pm 0.0785}$(13)	$0.9664_{\pm 0.0071}$(3)	$0.9873_{\pm 0.0092}$(13)	$0.7202_{\pm 0.0217}$(7)
HBOS	$0.9911_{\pm 0.0036}$(8)	$0.7848_{\pm 0.0621}$(12)	$0.8342_{\pm 0.0407}$(6)	$0.7334_{\pm 0.0519}$(18)	$0.9975_{\pm 0.0045}$(8)	$0.7339_{\pm 0.0201}$(2)
IForest	$0.9942_{\pm 0.0028}$(4)	$0.7855_{\pm 0.0429}$(11)	$0.8348_{\pm 0.0613}$(5)	$0.9116_{\pm 0.0189}$(13)	$0.9991_{\pm 0.0019}$(3)	$0.7272_{\pm 0.0275}$(5)
KDE	$0.9946_{\pm 0.0025}$(2)	$0.7622_{\pm 0.0454}$(13)	$0.7942_{\pm 0.0540}$(14)	$0.9520_{\pm 0.0105}$(8)	$0.9988_{\pm 0.0025}$(5)	$0.7208_{\pm 0.0226}$(6)
KPCA	$0.1983_{\pm 0.0825}$(21)	$0.3975_{\pm 0.1372}$(20)	$0.4379_{\pm 0.1895}$(21)	$0.4606_{\pm 0.0632}$(21)	$0.4137_{\pm 0.2537}$(20)	$0.4976_{\pm 0.0274}$(21)
LMDD	$0.7050_{\pm 0.1609}$(17)	$0.3743_{\pm 0.1793}$(22)	$0.5742_{\pm 0.1757}$(20)	$0.6665_{\pm 0.1000}$(19)	$0.9759_{\pm 0.0186}$(17)	$0.6293_{\pm 0.0860}$(18)
LOF	$0.4482_{\pm 0.0215}$(19)	$0.8048_{\pm 0.0784}$(9)	$0.8211_{\pm 0.0761}$(8)	$0.8665_{\pm 0.0479}$(14)	$0.9988_{\pm 0.0025}$(5)	$0.6348_{\pm 0.0227}$(17)
LUNAR	$0.9902_{\pm 0.0046}$(10)	$0.8816_{\pm 0.0359}$(1)	$0.7782_{\pm 0.0579}$(15)	$0.9781_{\pm 0.0085}$(1)	$0.9906_{\pm 0.0112}$(11)	$0.7296_{\pm 0.0286}$(4)
MCD	$0.9850_{\pm 0.0084}$(13)	$0.7869_{\pm 0.0347}$(10)	$0.8088_{\pm 0.0621}$(11)	$0.9616_{\pm 0.0092}$(4)	$0.9861_{\pm 0.0140}$(14)	$0.7158_{\pm 0.0222}$(8)
MO_GAAL	$0.0117_{\pm 0.0142}$(22)	$0.3744_{\pm 0.2277}$(21)	$0.3695_{\pm 0.1895}$(22)	$0.3329_{\pm 0.0612}$(22)	$0.0085_{\pm 0.0148}$(22)	$0.3319_{\pm 0.0499}$(23)
OCSVM	$0.9945_{\pm 0.0025}$(3)	$0.4470_{\pm 0.1473}$(19)	$0.8088_{\pm 0.0630}$(11)	$0.8405_{\pm 0.0192}$(15)	$0.9988_{\pm 0.0025}$(5)	$0.6797_{\pm 0.0246}$(12)
PCA	$0.9495_{\pm 0.0188}$(15)	$0.6551_{\pm 0.0838}$(18)	$0.8120_{\pm 0.0635}$(9)	$0.7864_{\pm 0.0215}$(16)	$0.9962_{\pm 0.0058}$(9)	$0.6377_{\pm 0.0241}$(15)
QMCD	$0.4800_{\pm 0.1746}$(18)	$0.8423_{\pm 0.0490}$(6)	$0.7150_{\pm 0.1246}$(18)	$0.5509_{\pm 0.0455}$(20)	$0.0884_{\pm 0.1204}$(21)	$0.7417_{\pm 0.0239}$(1)
SO_GAAL	$0.0110_{\pm 0.0116}$(23)	$0.3392_{\pm 0.1817}$(23)	$0.3134_{\pm 0.1751}$(23)	$0.2823_{\pm 0.0611}$(23)	$0.0049_{\pm 0.0108}$(23)	$0.3350_{\pm 0.0491}$(22)
Sampling	$0.9934_{\pm 0.0033}$(6)	$0.8696_{\pm 0.0480}$(4)	$0.7645_{\pm 0.0976}$(17)	$0.9395_{\pm 0.0256}$(10)	$0.9780_{\pm 0.0359}$(16)	$0.7115_{\pm 0.0369}$(9)
TCCM	$0.9677_{\pm 0.0195}$(14)	$0.8334_{\pm 0.0914}$(7)	$0.8394_{\pm 0.0708}$(4)	$0.9400_{\pm 0.0158}$(9)	$1.0000_{\pm 0.0000}$(1)	$0.5562_{\pm 0.1001}$(20)
VAE	$0.9903_{\pm 0.0045}$(9)	$0.7148_{\pm 0.0655}$(15)	$0.8600_{\pm 0.0425}$(1)	$0.9206_{\pm 0.0155}$(12)	$0.9932_{\pm 0.0068}$(10)	$0.6714_{\pm 0.0323}$(13)

TABLE XVII: ROC–AUC, small datasets (2/2).

Model	37_Stamps	39_vertebral	42_WBC	43_WDBC	45_wine	46_WPBC
ABOD	$0.6954_{\pm 0.0525}$(19)	$0.4214_{\pm 0.0847}$(18)	$0.9297_{\pm 0.0301}$(18)	$0.9143_{\pm 0.0504}$(18)	$0.5828_{\pm 0.2278}$(20)	$0.4897_{\pm 0.0563}$(19)
AutoEncoder	$0.9217_{\pm 0.0398}$(10)	$0.4644_{\pm 0.1041}$(12)	$0.9841_{\pm 0.0220}$(12)	$0.9937_{\pm 0.0055}$(2)	$0.9364_{\pm 0.0499}$(6)	$0.5178_{\pm 0.0481}$(11)
CBLOF	$0.9457_{\pm 0.0235}$(4)	$0.4711_{\pm 0.0742}$(10)	$0.9961_{\pm 0.0039}$(2)	$0.9860_{\pm 0.0099}$(14)	$0.9517_{\pm 0.0376}$(2)	$0.5280_{\pm 0.0407}$(9)
DIF	$0.9426_{\pm 0.0215}$(5)	$0.5781_{\pm 0.0895}$(3)	$0.7821_{\pm 0.0531}$(19)	$0.7227_{\pm 0.0519}$(19)	$0.7978_{\pm 0.1078}$(17)	$0.4971_{\pm 0.0485}$(18)
DiffInt	$0.9516_{\pm 0.0116}$(1)	$0.5274_{\pm 0.0429}$(4)	$0.9973_{\pm 0.0050}$(1)	$0.9741_{\pm 0.0076}$(16)	$0.8852_{\pm 0.0594}$(13)	$0.5652_{\pm 0.0222}$(1)
ECOD	$0.9157_{\pm 0.0244}$(13)	$0.4645_{\pm 0.0796}$(11)	$0.9953_{\pm 0.0046}$(6)	$0.9870_{\pm 0.0068}$(11)	$0.8484_{\pm 0.0960}$(14)	$0.5029_{\pm 0.0422}$(17)
GMM	$0.9270_{\pm 0.0226}$(9)	$0.4909_{\pm 0.0550}$(8)	$0.9767_{\pm 0.0198}$(16)	$0.9882_{\pm 0.0093}$(9)	$0.9677_{\pm 0.0268}$(1)	$0.5141_{\pm 0.0507}$(12)
HBOS	$0.9375_{\pm 0.0146}$(7)	$0.3982_{\pm 0.0437}$(21)	$0.9896_{\pm 0.0093}$(9)	$0.9890_{\pm 0.0063}$(7)	$0.9296_{\pm 0.0419}$(9)	$0.5483_{\pm 0.0532}$(5)
IForest	$0.9485_{\pm 0.0202}$(2)	$0.4426_{\pm 0.1017}$(15)	$0.9948_{\pm 0.0049}$(7)	$0.9903_{\pm 0.0056}$(4)	$0.9393_{\pm 0.0441}$(5)	$0.5287_{\pm 0.0451}$(8)
KDE	$0.9309_{\pm 0.0174}$(8)	$0.4096_{\pm 0.0856}$(20)	$0.9960_{\pm 0.0045}$(3)	$0.9864_{\pm 0.0075}$(13)	$0.9336_{\pm 0.0339}$(7)	$0.5603_{\pm 0.0511}$(2)
KPCA	$0.4111_{\pm 0.1136}$(21)	$0.5204_{\pm 0.0619}$(6)	$0.3710_{\pm 0.2036}$(20)	$0.2678_{\pm 0.0890}$(21)	$0.3955_{\pm 0.2631}$(21)	$0.4644_{\pm 0.0961}$(21)
LMDD	$0.4454_{\pm 0.0266}$(20)	$0.4355_{\pm 0.0811}$(16)	$0.9769_{\pm 0.0615}$(15)	$0.9941_{\pm 0.0033}$(1)	$0.6987_{\pm 0.1469}$(18)	$0.5125_{\pm 0.0619}$(14)
LOF	$0.7229_{\pm 0.1204}$(18)	$0.3515_{\pm 0.0717}$(23)	$0.9759_{\pm 0.0198}$(17)	$0.9871_{\pm 0.0088}$(10)	$0.9115_{\pm 0.0592}$(12)	$0.5326_{\pm 0.0635}$(7)
LUNAR	$0.9473_{\pm 0.0248}$(3)	$0.4484_{\pm 0.0817}$(13)	$0.9794_{\pm 0.0237}$(14)	$0.9935_{\pm 0.0055}$(3)	$0.9478_{\pm 0.0439}$(3)	$0.5536_{\pm 0.0417}$(3)
MCD	$0.8368_{\pm 0.0255}$(17)	$0.4872_{\pm 0.0529}$(9)	$0.9807_{\pm 0.0230}$(13)	$0.9682_{\pm 0.0141}$(17)	$0.9444_{\pm 0.0521}$(4)	$0.5277_{\pm 0.0682}$(10)
MO_GAAL	$0.2743_{\pm 0.1774}$(22)	$0.6858_{\pm 0.0735}$(1)	$0.0050_{\pm 0.0063}$(23)	$0.0074_{\pm 0.0074}$(22)	$0.0320_{\pm 0.0388}$(22)	$0.4373_{\pm 0.0912}$(22)
OCSVM	$0.9027_{\pm 0.0260}$(14)	$0.4294_{\pm 0.0843}$(17)	$0.9960_{\pm 0.0045}$(3)	$0.9887_{\pm 0.0057}$(8)	$0.8410_{\pm 0.0944}$(16)	$0.5137_{\pm 0.0344}$(13)
PCA	$0.9197_{\pm 0.0195}$(12)	$0.3973_{\pm 0.0598}$(22)	$0.9907_{\pm 0.0091}$(8)	$0.9868_{\pm 0.0058}$(12)	$0.8450_{\pm 0.0888}$(15)	$0.5096_{\pm 0.0543}$(15)
QMCD	$0.9025_{\pm 0.0571}$(15)	$0.4114_{\pm 0.0871}$(19)	$0.3578_{\pm 0.1567}$(21)	$0.5860_{\pm 0.1903}$(20)	$0.6633_{\pm 0.1655}$(19)	$0.5343_{\pm 0.0571}$(6)
SO_GAAL	$0.2334_{\pm 0.1414}$(23)	$0.6801_{\pm 0.0693}$(2)	$0.0056_{\pm 0.0065}$(22)	$0.0060_{\pm 0.0051}$(23)	$0.0248_{\pm 0.0276}$(23)	$0.4272_{\pm 0.0797}$(23)
Sampling	$0.9210_{\pm 0.0424}$(11)	$0.4449_{\pm 0.1259}$(14)	$0.9871_{\pm 0.0271}$(11)	$0.9817_{\pm 0.0095}$(15)	$0.9328_{\pm 0.0487}$(8)	$0.5505_{\pm 0.0742}$(4)
TCCM	$0.8880_{\pm 0.0393}$(16)	$0.5183_{\pm 0.0730}$(7)	$0.9960_{\pm 0.0063}$(3)	$0.9899_{\pm 0.0077}$(5)	$0.9261_{\pm 0.0498}$(10)	$0.4841_{\pm 0.0688}$(20)
VAE	$0.9404_{\pm 0.0175}$(6)	$0.5244_{\pm 0.0725}$(5)	$0.9873_{\pm 0.0166}$(10)	$0.9897_{\pm 0.0076}$(6)	$0.9259_{\pm 0.0335}$(11)	$0.5083_{\pm 0.0451}$(16)

TABLE XVIII: ROC–AUC, medium datasets (1/3).

Model	2_annthyroid	6_cardio	7_Cardiotocography	12_fault	19_landsat	20_letter
ABOD	$0.7439_{\pm 0.0124}(13)$	$0.5877_{\pm 0.0275}(19)$	$0.4917_{\pm 0.0184}(21)$	$0.6928_{\pm 0.0110}(10)$	$0.5164_{\pm 0.0127}(17)$	$0.8652_{\pm 0.0149}(3)$
AutoEncoder	$0.8612_{\pm 0.0192}(6)$	$0.9307_{\pm 0.0271}(12)$	$0.7202_{\pm 0.0335}(12)$	$0.7386_{\pm 0.0188}(4)$	$0.5773_{\pm 0.0111}(13)$	$0.8135_{\pm 0.0340}(7)$
CBLOF	$0.6578_{\pm 0.0179}(17)$	$0.9546_{\pm 0.0043}(7)$	$0.7815_{\pm 0.0315}(8)$	$0.7094_{\pm 0.0358}(8)$	$0.6907_{\pm 0.0179}(3)$	$0.7755_{\pm 0.0370}(9)$
DIF	$0.8025_{\pm 0.0282}(10)$	$0.9570_{\pm 0.0075}(6)$	$0.6877_{\pm 0.0171}(15)$	$0.7386_{\pm 0.0145}(4)$	$0.5947_{\pm 0.0167}(11)$	$0.6661_{\pm 0.0575}(12)$
DiffInt	$0.8330_{\pm 0.0094}(8)$	$0.9721_{\pm 0.0050}(2)$	$0.8023_{\pm 0.0122}(6)$	$0.7976_{\pm 0.0044}(2)$	$0.6540_{\pm 0.0045}(5)$	$0.9006_{\pm 0.0102}(2)$
ECOD	$0.8249_{\pm 0.0099}(9)$	$0.9519_{\pm 0.0047}(10)$	$0.8322_{\pm 0.0096}(3)$	$0.4919_{\pm 0.0126}(20)$	$0.3959_{\pm 0.0111}(21)$	$0.5871_{\pm 0.0418}(17)$
GMM	$0.8412_{\pm 0.0270}(7)$	$0.9448_{\pm 0.0075}(11)$	$0.7507_{\pm 0.0134}(10)$	$0.7024_{\pm 0.0160}(9)$	$0.4967_{\pm 0.0096}(19)$	$0.8473_{\pm 0.0350}(5)$
HBOS	$0.7367_{\pm 0.0502}(14)$	$0.8513_{\pm 0.0151}(15)$	$0.6938_{\pm 0.0249}(14)$	$0.6047_{\pm 0.0649}(15)$	$0.6991_{\pm 0.0124}(2)$	$0.5934_{\pm 0.0266}(16)$
IForest	$0.9183_{\pm 0.0069}(2)$	$0.9524_{\pm 0.0094}(8)$	$0.8110_{\pm 0.0260}(5)$	$0.6565_{\pm 0.0253}(11)$	$0.6003_{\pm 0.0210}(10)$	$0.6365_{\pm 0.0459}(13)$
KDE	$0.5866_{\pm 0.0150}(21)$	$0.9624_{\pm 0.0015}(5)$	$0.8378_{\pm 0.0079}(2)$	$0.7583_{\pm 0.0132}(3)$	$0.5658_{\pm 0.0105}(15)$	$0.7064_{\pm 0.0385}(11)$
KPCA	$0.7473_{\pm 0.0161}(12)$	$0.4696_{\pm 0.0601}(21)$	$0.4918_{\pm 0.0441}(20)$	$0.5139_{\pm 0.0176}(19)$	$0.5017_{\pm 0.0162}(18)$	$0.5057_{\pm 0.0391}(19)$
LMDD	$0.5926_{\pm 0.0166}(20)$	$0.5774_{\pm 0.0236}(20)$	$0.7130_{\pm 0.0214}(13)$	$0.5351_{\pm 0.0282}(18)$	$0.4150_{\pm 0.0511}(20)$	$0.4388_{\pm 0.0308}(21)$
LOF	$0.7080_{\pm 0.0130}(15)$	$0.7194_{\pm 0.0416}(17)$	$0.5736_{\pm 0.0207}(18)$	$0.6000_{\pm 0.0126}(16)$	$0.5340_{\pm 0.0131}(16)$	$0.8203_{\pm 0.0272}(6)$
LUNAR	$0.8971_{\pm 0.0168}(3)$	$0.9631_{\pm 0.0038}(4)$	$0.7980_{\pm 0.0203}(7)$	$0.8110_{\pm 0.0120}(\mathbf{1})$	$0.7870_{\pm 0.0066}(\mathbf{1})$	$0.9319_{\pm 0.0166}(\mathbf{1})$
MCD	$0.9216_{\pm 0.0065}(\mathbf{1})$	$0.8603_{\pm 0.0172}(14)$	$0.6641_{\pm 0.0230}(16)$	$0.6222_{\pm 0.0350}(13)$	$0.6155_{\pm 0.0053}(8)$	$0.8111_{\pm 0.0230}(8)$
MO_GAAL	$0.5521_{\pm 0.0633}(22)$	$0.2338_{\pm 0.1069}(23)$	$0.4305_{\pm 0.1216}(22)$	$0.4666_{\pm 0.0522}(22)$	$0.6447_{\pm 0.0239}(7)$	$0.4041_{\pm 0.0506}(22)$
OCSVM	$0.5951_{\pm 0.0141}(19)$	$0.9521_{\pm 0.0034}(9)$	$0.8493_{\pm 0.0079}(\mathbf{1})$	$0.5551_{\pm 0.0202}(17)$	$0.3788_{\pm 0.0105}(22)$	$0.5003_{\pm 0.0482}(20)$
PCA	$0.6734_{\pm 0.0175}(16)$	$0.7263_{\pm 0.2385}(16)$	$0.5786_{\pm 0.1266}(17)$	$0.4850_{\pm 0.0191}(21)$	$0.3652_{\pm 0.0112}(23)$	$0.5287_{\pm 0.0401}(18)$
QMCD	$0.7681_{\pm 0.0130}(11)$	$0.7117_{\pm 0.0402}(18)$	$0.5658_{\pm 0.0256}(19)$	$0.6437_{\pm 0.0409}(12)$	$0.6834_{\pm 0.0094}(4)$	$0.6183_{\pm 0.0358}(15)$
SO_GAAL	$0.5461_{\pm 0.0616}(23)$	$0.2399_{\pm 0.1106}(22)$	$0.4256_{\pm 0.1214}(23)$	$0.4652_{\pm 0.0450}(23)$	$0.5759_{\pm 0.0248}(14)$	$0.4014_{\pm 0.0512}(23)$
Sampling	$0.6445_{\pm 0.0226}(18)$	$0.8934_{\pm 0.0399}(13)$	$0.7324_{\pm 0.0492}(11)$	$0.7165_{\pm 0.0336}(7)$	$0.6522_{\pm 0.0704}(6)$	$0.7229_{\pm 0.0388}(10)$
TCCM	$0.8840_{\pm 0.1138}(4)$	$0.9775_{\pm 0.0094}(\mathbf{1})$	$0.7653_{\pm 0.0441}(9)$	$0.7362_{\pm 0.0257}(6)$	$0.6044_{\pm 0.0186}(9)$	$0.8574_{\pm 0.0287}(4)$
VAE	$0.8821_{\pm 0.0227}(5)$	$0.9673_{\pm 0.0092}(3)$	$0.8153_{\pm 0.0201}(4)$	$0.6082_{\pm 0.0315}(14)$	$0.5807_{\pm 0.0081}(12)$	$0.6259_{\pm 0.0420}(14)$

TABLE XIX: ROC–AUC, medium datasets (2/3).

Model	27_PageBlocks	28_pendigits	30_satellite	31_satimage-2	38_thyroid	40_vowels
ABOD	$0.7166_{\pm 0.0251}(20)$	$0.6559_{\pm 0.0303}(18)$	$0.5619_{\pm 0.0148}(20)$	$0.7878_{\pm 0.0471}(18)$	$0.9111_{\pm 0.0147}(16)$	$0.9440_{\pm 0.0194}(6)$
AutoEncoder	$0.9434_{\pm 0.0122}(4)$	$0.9805_{\pm 0.0078}(5)$	$0.8047_{\pm 0.0102}(8)$	$0.9975_{\pm 0.0020}(7)$	$0.9816_{\pm 0.0080}(7)$	$0.9460_{\pm 0.0187}(5)$
CBLOF	$0.8792_{\pm 0.0185}(14)$	$0.9760_{\pm 0.0054}(7)$	$0.8575_{\pm 0.0259}(4)$	$0.9985_{\pm 0.0019}(3)$	$0.9354_{\pm 0.0105}(14)$	$0.9343_{\pm 0.0234}(7)$
DIF	$0.9354_{\pm 0.0115}(5)$	$0.9815_{\pm 0.0070}(4)$	$0.7886_{\pm 0.0133}(12)$	$0.9979_{\pm 0.0017}(4)$	$0.9838_{\pm 0.0039}(6)$	$0.8136_{\pm 0.0444}(10)$
DiffInt	$0.9500_{\pm 0.0043}(2)$	$0.9966_{\pm 0.0011}(2)$	$0.8416_{\pm 0.0046}(5)$	$0.9993_{\pm 0.0010}(\mathbf{1})$	$0.9770_{\pm 0.0047}(8)$	$0.9808_{\pm 0.0068}(2)$
ECOD	$0.9253_{\pm 0.0030}(7)$	$0.9329_{\pm 0.0094}(14)$	$0.6155_{\pm 0.0115}(16)$	$0.9749_{\pm 0.0169}(17)$	$0.9859_{\pm 0.0025}(4)$	$0.6123_{\pm 0.0509}(18)$
GMM	$0.9577_{\pm 0.0054}(\mathbf{1})$	$0.8485_{\pm 0.0120}(16)$	$0.8028_{\pm 0.0065}(10)$	$0.9958_{\pm 0.0013}(10)$	$0.9760_{\pm 0.0050}(10)$	$0.9480_{\pm 0.0177}(4)$
HBOS	$0.8133_{\pm 0.0120}(16)$	$0.9377_{\pm 0.0095}(12)$	$0.8624_{\pm 0.0077}(2)$	$0.9841_{\pm 0.0135}(15)$	$0.9877_{\pm 0.0039}(3)$	$0.7046_{\pm 0.0568}(14)$
IForest	$0.9200_{\pm 0.0099}(8)$	$0.9695_{\pm 0.0051}(8)$	$0.7967_{\pm 0.0151}(11)$	$0.9951_{\pm 0.0048}(11)$	$0.9878_{\pm 0.0034}(2)$	$0.7874_{\pm 0.0307}(11)$
KDE	$0.8868_{\pm 0.0099}(13)$	$0.9918_{\pm 0.0025}(3)$	$0.7678_{\pm 0.0099}(13)$	$0.9977_{\pm 0.0031}(6)$	$0.8526_{\pm 0.0155}(18)$	$0.7116_{\pm 0.0410}(13)$
KPCA	$0.6020_{\pm 0.0179}(21)$	$0.5025_{\pm 0.0867}(20)$	$0.4778_{\pm 0.0156}(21)$	$0.4588_{\pm 0.1931}(21)$	$0.3309_{\pm 0.0614}(23)$	$0.4887_{\pm 0.0458}(21)$
LMDD	$0.7235_{\pm 0.0263}(19)$	$0.6073_{\pm 0.0723}(19)$	$0.4468_{\pm 0.0678}(22)$	$0.5015_{\pm 0.1118}(20)$	$0.7074_{\pm 0.0709}(20)$	$0.5107_{\pm 0.0551}(20)$
LOF	$0.7654_{\pm 0.0132}(17)$	$0.4669_{\pm 0.0438}(21)$	$0.5634_{\pm 0.0141}(19)$	$0.4551_{\pm 0.2150}(22)$	$0.9417_{\pm 0.0230}(13)$	$0.9497_{\pm 0.0148}(3)$
LUNAR	$0.9353_{\pm 0.0114}(6)$	$0.9994_{\pm 0.0005}(\mathbf{1})$	$0.8803_{\pm 0.0073}(\mathbf{1})$	$0.9978_{\pm 0.0024}(5)$	$0.9770_{\pm 0.0053}(8)$	$0.9887_{\pm 0.0052}(\mathbf{1})$
MCD	$0.9195_{\pm 0.0075}(9)$	$0.8298_{\pm 0.0103}(17)$	$0.8093_{\pm 0.0073}(7)$	$0.9960_{\pm 0.0010}(9)$	$0.9852_{\pm 0.0029}(5)$	$0.7523_{\pm 0.0651}(12)$
MO_GAAL	$0.2884_{\pm 0.0932}(23)$	$0.2023_{\pm 0.0833}(22)$	$0.4177_{\pm 0.0347}(23)$	$0.2075_{\pm 0.1385}(23)$	$0.7068_{\pm 0.0820}(21)$	$0.1081_{\pm 0.0740}(22)$
OCSVM	$0.8962_{\pm 0.0092}(12)$	$0.9643_{\pm 0.0051}(9)$	$0.6537_{\pm 0.0126}(15)$	$0.9859_{\pm 0.0140}(14)$	$0.8610_{\pm 0.0151}(17)$	$0.5997_{\pm 0.0462}(19)$
PCA	$0.9061_{\pm 0.0075}(11)$	$0.9360_{\pm 0.0066}(13)$	$0.5999_{\pm 0.0117}(18)$	$0.9820_{\pm 0.0155}(16)$	$0.9571_{\pm 0.0053}(11)$	$0.6158_{\pm 0.0514}(17)$
QMCD	$0.7648_{\pm 0.0253}(18)$	$0.9106_{\pm 0.0099}(15)$	$0.8605_{\pm 0.0072}(3)$	$0.9961_{\pm 0.0036}(8)$	$0.8329_{\pm 0.0760}(19)$	$0.7020_{\pm 0.0460}(15)$
SO_GAAL	$0.2970_{\pm 0.0840}(22)$	$0.1966_{\pm 0.0956}(23)$	$0.6032_{\pm 0.1301}(17)$	$0.7099_{\pm 0.2435}(19)$	$0.6946_{\pm 0.0803}(22)$	$0.0992_{\pm 0.0719}(23)$
Sampling	$0.8629_{\pm 0.0352}(15)$	$0.9379_{\pm 0.0366}(11)$	$0.8141_{\pm 0.0294}(6)$	$0.9946_{\pm 0.0072}(12)$	$0.9239_{\pm 0.0183}(15)$	$0.8896_{\pm 0.0486}(9)$
TCCM	$0.9152_{\pm 0.0262}(10)$	$0.9774_{\pm 0.0198}(6)$	$0.8029_{\pm 0.0111}(9)$	$0.9989_{\pm 0.0014}(2)$	$0.9443_{\pm 0.0065}(12)$	$0.9339_{\pm 0.0172}(8)$
VAE	$0.9477_{\pm 0.0099}(3)$	$0.9454_{\pm 0.0062}(10)$	$0.7602_{\pm 0.0093}(14)$	$0.9872_{\pm 0.0081}(13)$	$0.9883_{\pm 0.0027}(\mathbf{1})$	$0.6742_{\pm 0.0538}(16)$

TABLE XX: ROC–AUC, medium datasets (3/3).

Model	41_Waveform	44_Wilt	47_yeast
ABOD	$0.6265_{\pm 0.0327}(15)$	$0.5544_{\pm 0.0226}(9)$	$0.4188_{\pm 0.0212}(18)$
AutoEncoder	$0.6848_{\pm 0.0565}(12)$	$0.5556_{\pm 0.0760}(8)$	$0.4561_{\pm 0.0156}(9)$
CBLOF	$0.7546_{\pm 0.0522}(3)$	$0.3755_{\pm 0.0282}(19)$	$0.4777_{\pm 0.0209}(6)$
DIF	$0.7470_{\pm 0.0248}(5)$	$0.4179_{\pm 0.0614}(16)$	$0.3969_{\pm 0.0183}(23)$
DiffInt	$0.7678_{\pm 0.0178}(\mathbf{1})$	$0.6789_{\pm 0.0272}(5)$	$0.4926_{\pm 0.0163}(5)$
ECOD	$0.6140_{\pm 0.0352}(16)$	$0.3994_{\pm 0.0286}(18)$	$0.4492_{\pm 0.0167}(12)$
GMM	$0.5790_{\pm 0.0356}(17)$	$0.7295_{\pm 0.0295}(2)$	$0.4374_{\pm 0.0190}(14)$
HBOS	$0.6973_{\pm 0.0316}(11)$	$0.4051_{\pm 0.0995}(17)$	$0.4213_{\pm 0.0180}(17)$
IForest	$0.7332_{\pm 0.0412}(8)$	$0.4687_{\pm 0.0343}(10)$	$0.4106_{\pm 0.0163}(19)$
KDE	$0.7661_{\pm 0.0269}(2)$	$0.3569_{\pm 0.0299}(21)$	$0.4034_{\pm 0.0169}(20)$
KPCA	$0.5138_{\pm 0.0243}(21)$	$0.6498_{\pm 0.0243}(6)$	$0.5649_{\pm 0.0142}(\mathbf{1})$
LMDD	$0.5173_{\pm 0.0232}(20)$	$0.4192_{\pm 0.0570}(15)$	$0.4642_{\pm 0.0230}(7)$
LOF	$0.7395_{\pm 0.0396}(7)$	$0.4265_{\pm 0.0151}(13)$	$0.4327_{\pm 0.0215}(15)$
LUNAR	$0.7528_{\pm 0.0484}(4)$	$0.5917_{\pm 0.0676}(7)$	$0.4477_{\pm 0.0207}(13)$
MCD	$0.5783_{\pm 0.0365}(18)$	$0.8608_{\pm 0.0102}(\mathbf{1})$	$0.4541_{\pm 0.0228}(10)$
MO_GAAL	$0.3040_{\pm 0.0835}(23)$	$0.6966_{\pm 0.0301}(3)$	$0.5599_{\pm 0.0385}(3)$
OCSVM	$0.5662_{\pm 0.0422}(19)$	$0.3574_{\pm 0.0313}(20)$	$0.4260_{\pm 0.0158}(16)$
PCA	$0.6432_{\pm 0.0373}(13)$	$0.4404_{\pm 0.0234}(12)$	$0.3982_{\pm 0.0167}(22)$
QMCD	$0.7448_{\pm 0.0245}(6)$	$0.3565_{\pm 0.0351}(23)$	$0.3991_{\pm 0.0196}(21)$
SO_GAAL	$0.3466_{\pm 0.0815}(22)$	$0.6906_{\pm 0.0275}(4)$	$0.5642_{\pm 0.0432}(2)$
Sampling	$0.7022_{\pm 0.0578}(10)$	$0.4217_{\pm 0.0380}(14)$	$0.4602_{\pm 0.0513}(8)$
TCCM	$0.6406_{\pm 0.0827}(14)$	$0.3567_{\pm 0.0623}(22)$	$0.5201_{\pm 0.0587}(4)$
VAE	$0.7081_{\pm 0.0266}(9)$	$0.4547_{\pm 0.0278}(11)$	$0.4500_{\pm 0.0162}(11)$

TABLE XXI: ROC–AUC, large datasets (1/2).

Model	1_ALOI	8_celeba	10_cover	11_donors	13_fraud	16_http
ABOD	$0.7311_{\pm 0.0082}(3)$	$0.4706_{\pm 0.0053}(19)$	$0.7391_{\pm 0.0072}(16)$	$0.4036_{\pm 0.0014}(16)$	$0.8889_{\pm 0.0172}(17)$	$0.7510_{\pm 0.0017}(16)$
AutoEncoder	$0.5554_{\pm 0.0126}(6)$	$0.6988_{\pm 0.0311}(11)$	$0.9843_{\pm 0.0044}(2)$	$0.9359_{\pm 0.0100}(3)$	$0.9592_{\pm 0.0070}(4)$	$0.9988_{\pm 0.0012}(6)$
CBLOF	$0.5449_{\pm 0.0101}(11)$	$0.6790_{\pm 0.0629}(12)$	$0.9245_{\pm 0.0174}(9)$	$0.9159_{\pm 0.0156}(6)$	$0.9535_{\pm 0.0052}(9)$	$0.9949_{\pm 0.0001}(12)$
DIF	$0.5498_{\pm 0.0103}(9)$	$0.6636_{\pm 0.0155}(14)$	$0.9787_{\pm 0.0040}(3)$	$0.9118_{\pm 0.0101}(8)$	$0.9528_{\pm 0.0078}(10)$	$0.9935_{\pm 0.0002}(14)$
DiffInt	$0.5857_{\pm 0.0067}(4)$	$0.7463_{\pm 0.0386}(8)$	$0.9949_{\pm 0.0015}(\mathbf{1})$	$1.0000_{\pm 0.0000}(\mathbf{1})$	$0.9596_{\pm 0.0025}(3)$	$0.9998_{\pm 0.0022}(\mathbf{1})$
ECOD	$0.5326_{\pm 0.0086}(16)$	$0.7664_{\pm 0.0030}(7)$	$0.9386_{\pm 0.0019}(6)$	$0.9215_{\pm 0.0005}(5)$	$0.9525_{\pm 0.0057}(11)$	$0.9880_{\pm 0.0003}(15)$
GMM	$0.5601_{\pm 0.0100}(5)$	$0.8076_{\pm 0.0032}(3)$	$0.9505_{\pm 0.0012}(5)$	$0.9255_{\pm 0.0004}(4)$	$0.9597_{\pm 0.0053}(2)$	$0.9994_{\pm 0.0000}(4)$
HBOS	$0.5323_{\pm 0.0091}(17)$	$0.7685_{\pm 0.0031}(6)$	$0.7880_{\pm 0.0041}(15)$	$0.7922_{\pm 0.0016}(13)$	$0.9551_{\pm 0.0060}(7)$	$0.9961_{\pm 0.0014}(8)$
IForest	$0.5382_{\pm 0.0112}(12)$	$0.7127_{\pm 0.0194}(9)$	$0.8456_{\pm 0.0235}(12)$	$0.9124_{\pm 0.0222}(7)$	$0.9513_{\pm 0.0063}(12)$	$0.9938_{\pm 0.0029}(13)$
KDE	$0.5367_{\pm 0.0103}(14)$	$0.6675_{\pm 0.0039}(13)$	$0.8716_{\pm 0.0019}(10)$	-	-	-
KPCA	$0.5327_{\pm 0.0114}(15)$	-	-	-	-	-
LMDD	$0.5322_{\pm 0.0124}(18)$	-	-	-	-	-
LOF	$0.7347_{\pm 0.0109}(2)$	$0.4171_{\pm 0.0091}(20)$	$0.5547_{\pm 0.0107}(18)$	$0.6111_{\pm 0.0036}(15)$	$0.4976_{\pm 0.0086}(18)$	$0.3929_{\pm 0.0052}(18)$
LUNAR	$0.7389_{\pm 0.0114}(\mathbf{1})$	$0.6335_{\pm 0.0069}(15)$	-	-	$0.9658_{\pm 0.0057}(\mathbf{1})$	$0.9953_{\pm 0.0037}(10)$
MCD	$0.5262_{\pm 0.0088}(20)$	$0.8246_{\pm 0.0175}(2)$	$0.7027_{\pm 0.0025}(17)$	$0.8202_{\pm 0.0993}(12)$	$0.9232_{\pm 0.0077}(16)$	$0.9995_{\pm 0.0000}(3)$
MO_GAAL	$0.4740_{\pm 0.0121}(23)$	$0.2365_{\pm 0.0809}(21)$	-	-	-	-
OCSVM	$0.5320_{\pm 0.0086}(19)$	$0.7100_{\pm 0.0033}(10)$	$0.9251_{\pm 0.0020}(8)$	-	$0.9501_{\pm 0.0072}(13)$	-
PCA	$0.5485_{\pm 0.0107}(10)$	$0.7851_{\pm 0.0032}(4)$	$0.9337_{\pm 0.0010}(7)$	$0.8240_{\pm 0.0011}(11)$	$0.9546_{\pm 0.0052}(8)$	$0.9961_{\pm 0.0001}(8)$
QMCD	$0.5258_{\pm 0.0096}(21)$	$0.5000_{\pm 0.0000}(17)$	$0.8215_{\pm 0.0030}(14)$	$0.7250_{\pm 0.0033}(14)$	$0.9575_{\pm 0.0059}(5)$	$0.9968_{\pm 0.0002}(7)$
SO_GAAL	$0.4770_{\pm 0.0113}(22)$	$0.4803_{\pm 0.0248}(18)$	$0.5161_{\pm 0.0088}(19)$	$0.3832_{\pm 0.1452}(17)$	$0.0751_{\pm 0.0216}(19)$	$0.4447_{\pm 0.1537}(17)$
Sampling	$0.5376_{\pm 0.0108}(13)$	$0.6126_{\pm 0.1041}(16)$	$0.8391_{\pm 0.0579}(13)$	$0.8793_{\pm 0.0257}(9)$	$0.9472_{\pm 0.0106}(14)$	$0.9952_{\pm 0.0014}(11)$
TCCM	$0.5501_{\pm 0.0163}(8)$	$0.8308_{\pm 0.0109}(\mathbf{1})$	$0.8705_{\pm 0.1019}(11)$	$0.9955_{\pm 0.0057}(2)$	$0.9404_{\pm 0.0136}(15)$	$0.9997_{\pm 0.0007}(2)$
VAE	$0.5541_{\pm 0.0084}(7)$	$0.7690_{\pm 0.0035}(5)$	$0.9745_{\pm 0.0009}(4)$	$0.8428_{\pm 0.0458}(10)$	$0.9574_{\pm 0.0045}(6)$	$0.9994_{\pm 0.0000}(4)$

TABLE XXII: ROC–AUC, large datasets (2/2).

Model	22_magic	23_mammography	32_shuttle	33_skin	34_smtp
ABOD	$0.7930_{\pm 0.0069}(7)$	$0.5424_{\pm 0.0121}(21)$	$0.6135_{\pm 0.0077}(20)$	$0.4064_{\pm 0.0026}(20)$	$0.8606_{\pm 0.0198}(9)$
AutoEncoder	$0.8211_{\pm 0.0111}(4)$	$0.9114_{\pm 0.0197}(2)$	$0.9968_{\pm 0.0010}(3)$	$0.7975_{\pm 0.0685}(9)$	$0.9285_{\pm 0.0433}(2)$
CBLOF	$0.8235_{\pm 0.0062}(3)$	$0.8238_{\pm 0.0188}(13)$	$0.9871_{\pm 0.0029}(15)$	$0.9178_{\pm 0.0035}(3)$	$0.8446_{\pm 0.0747}(14)$
DIF	$0.7822_{\pm 0.0136}(8)$	$0.8281_{\pm 0.0231}(12)$	$0.9905_{\pm 0.0023}(9)$	$0.8759_{\pm 0.0087}(8)$	$0.8490_{\pm 0.0617}(13)$
DiffInt	$0.8644_{\pm 0.0026}(\mathbf{1})$	$0.8803_{\pm 0.0097}(7)$	$0.9992_{\pm 0.0004}(2)$	$0.9909_{\pm 0.0021}(\mathbf{1})$	$0.8961_{\pm 0.0151}(6)$
ECOD	$0.6541_{\pm 0.0055}(19)$	$0.9115_{\pm 0.0163}(\mathbf{1})$	$0.9962_{\pm 0.0007}(5)$	$0.5421_{\pm 0.0011}(18)$	$0.8868_{\pm 0.0419}(7)$
GMM	$0.8040_{\pm 0.0036}(5)$	$0.8814_{\pm 0.0198}(6)$	$0.9936_{\pm 0.0016}(8)$	$0.8885_{\pm 0.0004}(6)$	$0.8311_{\pm 0.0862}(16)$
HBOS	$0.7301_{\pm 0.0060}(13)$	$0.8706_{\pm 0.0169}(8)$	$0.9877_{\pm 0.0037}(13)$	$0.7722_{\pm 0.0011}(10)$	$0.8528_{\pm 0.0687}(12)$
IForest	$0.7764_{\pm 0.0148}(9)$	$0.8832_{\pm 0.0230}(4)$	$0.9965_{\pm 0.0008}(4)$	$0.8909_{\pm 0.0043}(5)$	$0.9061_{\pm 0.0377}(5)$
KDE	$0.7624_{\pm 0.0043}(11)$	$0.8581_{\pm 0.0153}(9)$	$0.9875_{\pm 0.0021}(14)$	$0.7324_{\pm 0.0012}(12)$	$0.8026_{\pm 0.0518}(17)$
KPCA	$0.5247_{\pm 0.0061}(20)$	$0.7860_{\pm 0.0135}(17)$	$0.9270_{\pm 0.0039}(17)$	-	-
LMDD	$0.5204_{\pm 0.0306}(21)$	$0.7440_{\pm 0.0772}(18)$	$0.6638_{\pm 0.1080}(19)$	-	$0.6184_{\pm 0.1197}(20)$
LOF	$0.6790_{\pm 0.0044}(17)$	$0.8018_{\pm 0.0235}(15)$	$0.5391_{\pm 0.0107}(21)$	$0.5472_{\pm 0.0032}(17)$	$0.9139_{\pm 0.0409}(4)$
LUNAR	$0.8554_{\pm 0.0047}(2)$	$0.8817_{\pm 0.0209}(5)$	$0.9997_{\pm 0.0002}(\mathbf{1})$	$0.9901_{\pm 0.0015}(2)$	$0.9280_{\pm 0.0266}(3)$
MCD	$0.7380_{\pm 0.0043}(12)$	$0.7199_{\pm 0.1363}(20)$	$0.9898_{\pm 0.0009}(11)$	$0.8844_{\pm 0.0004}(7)$	$0.9508_{\pm 0.0149}(\mathbf{1})$
MO_GAAL	$0.2031_{\pm 0.0117}(23)$	$0.1150_{\pm 0.0270}(22)$	$0.0289_{\pm 0.0036}(23)$	-	$0.4082_{\pm 0.0658}(22)$
OCSVM	$0.7030_{\pm 0.0048}(16)$	$0.8486_{\pm 0.0127}(11)$	$0.9869_{\pm 0.0022}(16)$	$0.7610_{\pm 0.0013}(11)$	$0.7982_{\pm 0.0538}(18)$
PCA	$0.6666_{\pm 0.0062}(18)$	$0.8923_{\pm 0.0151}(3)$	$0.9899_{\pm 0.0017}(10)$	$0.4149_{\pm 0.0026}(19)$	$0.8348_{\pm 0.0519}(15)$
QMCD	$0.7157_{\pm 0.0053}(14)$	$0.7313_{\pm 0.0276}(19)$	$0.7274_{\pm 0.0097}(18)$	$0.6219_{\pm 0.0010}(15)$	$0.7361_{\pm 0.0453}(19)$
SO_GAAL	$0.2044_{\pm 0.0145}(22)$	$0.1149_{\pm 0.0253}(23)$	$0.0297_{\pm 0.0039}(22)$	$0.5628_{\pm 0.2584}(16)$	$0.4270_{\pm 0.1116}(21)$
Sampling	$0.7934_{\pm 0.0237}(6)$	$0.8047_{\pm 0.0303}(14)$	$0.9892_{\pm 0.0030}(12)$	$0.9165_{\pm 0.0304}(4)$	$0.8701_{\pm 0.0589}(8)$
TCCM	$0.7748_{\pm 0.0216}(10)$	$0.8503_{\pm 0.0335}(10)$	$0.9961_{\pm 0.0007}(6)$	$0.6414_{\pm 0.2798}(13)$	$0.8591_{\pm 0.0747}(11)$
VAE	$0.7088_{\pm 0.0183}(15)$	$0.7924_{\pm 0.0272}(16)$	$0.9948_{\pm 0.0012}(7)$	$0.6267_{\pm 0.0018}(14)$	$0.8595_{\pm 0.0712}(10)$

TABLE XXIII: ROC–AUC, high-dimensional datasets (1/2).

Model	3_backdoor	5_campaign	9_census	17_InternetAds	24_mnist	25_musk
ABOD	$0.6392_{\pm 0.0091}(17)$	$0.6588_{\pm 0.0065}(16)$	$0.4026_{\pm 0.0026}(9)$	$0.6577_{\pm 0.0328}(14)$	$0.7120_{\pm 0.0145}(16)$	$0.2821_{\pm 0.0881}(20)$
AutoEncoder	$0.9331_{\pm 0.0084}(5)$	$0.8128_{\pm 0.0074}(\mathbf{1})$	-	$0.8806_{\pm 0.0147}(3)$	$0.9345_{\pm 0.0081}(\mathbf{1})$	$1.0000_{\pm 0.0000}(\mathbf{1})$
CBLOF	$0.8396_{\pm 0.0204}(12)$	$0.6743_{\pm 0.0125}(15)$	-	$0.7185_{\pm 0.0115}(7)$	$0.8727_{\pm 0.0109}(11)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
DIF	$0.9311_{\pm 0.0067}(6)$	$0.6776_{\pm 0.0116}(14)$	-	$0.6346_{\pm 0.0217}(16)$	$0.8818_{\pm 0.0097}(8)$	$0.9999_{\pm 0.0001}(11)$
DiffInt	$0.9613_{\pm 0.0043}(\mathbf{1})$	$0.8106_{\pm 0.0050}(2)$	$0.6273_{\pm 0.0007}(4)$	$0.7036_{\pm 0.0053}(11)$	$0.8950_{\pm 0.0074}(6)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
ECOD	$0.8567_{\pm 0.0077}(11)$	$0.7836_{\pm 0.0040}(8)$	-	$0.7079_{\pm 0.0116}(9)$	$0.7814_{\pm 0.0121}(14)$	$0.9984_{\pm 0.0008}(14)$
GMM	$0.9276_{\pm 0.0083}(7)$	$0.7973_{\pm 0.0040}(6)$	-	$0.9123_{\pm 0.0089}(\mathbf{1})$	$0.9240_{\pm 0.0080}(5)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
HBOS	$0.6620_{\pm 0.0077}(16)$	$0.8011_{\pm 0.0046}(5)$	-	$0.4937_{\pm 0.0488}(21)$	$0.6706_{\pm 0.0110}(17)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
IForest	$0.7612_{\pm 0.0352}(15)$	$0.7285_{\pm 0.0159}(10)$	$0.6185_{\pm 0.0120}(5)$	$0.4246_{\pm 0.0516}(23)$	$0.8645_{\pm 0.0181}(12)$	$0.9409_{\pm 0.0387}(16)$
KDE	$0.9386_{\pm 0.0068}(4)$	$0.6999_{\pm 0.0065}(12)$	-	$0.8524_{\pm 0.0118}(6)$	$0.9336_{\pm 0.0067}(2)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
KPCA	-	$0.4083_{\pm 0.0078}(23)$	-	$0.5284_{\pm 0.0289}(18)$	$0.5050_{\pm 0.0231}(19)$	$0.5021_{\pm 0.0694}(17)$
LMDD	-	$0.4491_{\pm 0.0581}(20)$	-	$0.6700_{\pm 0.0126}(13)$	$0.3897_{\pm 0.0343}(23)$	$0.3448_{\pm 0.0445}(19)$
LOF	$0.7669_{\pm 0.0131}(14)$	$0.5922_{\pm 0.0063}(18)$	$0.5308_{\pm 0.0024}(6)$	$0.6397_{\pm 0.0372}(15)$	$0.7276_{\pm 0.0241}(15)$	$0.3703_{\pm 0.0383}(18)$
LUNAR	$0.9539_{\pm 0.0064}(2)$	$0.7266_{\pm 0.0085}(11)$	-	$0.8835_{\pm 0.0133}(2)$	$0.9286_{\pm 0.0082}(4)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
MCD	$0.8659_{\pm 0.0879}(10)$	$0.7961_{\pm 0.0082}(7)$	$0.7426_{\pm 0.0106}(\mathbf{1})$	$0.5165_{\pm 0.0600}(19)$	$0.8790_{\pm 0.0245}(9)$	$0.9998_{\pm 0.0005}(12)$
MO_GAAL	$0.4091_{\pm 0.2470}(20)$	$0.4278_{\pm 0.0608}(22)$	-	$0.4862_{\pm 0.0435}(22)$	$0.4932_{\pm 0.1136}(22)$	$0.2547_{\pm 0.1279}(22)$
OCSVM	$0.8668_{\pm 0.0070}(9)$	$0.6903_{\pm 0.0052}(13)$	-	$0.7079_{\pm 0.0117}(9)$	$0.8840_{\pm 0.0109}(7)$	$0.9927_{\pm 0.0040}(15)$
PCA	$0.5000_{\pm 0.0000}(19)$	$0.5000_{\pm 0.0000}(19)$	$0.5000_{\pm 0.0000}(7)$	$0.6157_{\pm 0.0151}(17)$	$0.5000_{\pm 0.0000}(21)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
QMCD	$0.1021_{\pm 0.0065}(21)$	$0.8029_{\pm 0.0074}(4)$	$0.0000_{\pm 0.0019}(10)$	$0.8591_{\pm 0.0185}(5)$	$0.6071_{\pm 0.0088}(18)$	$0.0329_{\pm 0.0160}(23)$
SO_GAAL	$0.5055_{\pm 0.1733}(18)$	$0.4387_{\pm 0.0670}(21)$	$0.4850_{\pm 0.0353}(8)$	$0.5000_{\pm 0.0000}(20)$	$0.5021_{\pm 0.1107}(20)$	$0.2675_{\pm 0.1256}(21)$
Sampling	$0.8221_{\pm 0.0878}(13)$	$0.6508_{\pm 0.0233}(17)$	-	$0.7121_{\pm 0.0163}(8)$	$0.8177_{\pm 0.0664}(13)$	$0.9997_{\pm 0.0004}(13)$
TCCM	$0.9526_{\pm 0.0111}(3)$	$0.8052_{\pm 0.0177}(3)$	$0.6896_{\pm 0.0068}(3)$	$0.6984_{\pm 0.0109}(12)$	$0.8728_{\pm 0.0247}(10)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
VAE	$0.9129_{\pm 0.0046}(8)$	$0.7800_{\pm 0.0039}(9)$	$0.7048_{\pm 0.0021}(2)$	$0.8782_{\pm 0.0151}(4)$	$0.9332_{\pm 0.0083}(3)$	$1.0000_{\pm 0.0000}(\mathbf{1})$

TABLE XXIV: ROC–AUC, high-dimensional datasets (2/2).

Model	26_optdigits	35_SpamBase	36_speech	arrhythmia
ABOD	$0.4934_{\pm 0.0350}$(18)	$0.4517_{\pm 0.0131}$(20)	$0.6062_{\pm 0.0881}$(2)	$0.7597_{\pm 0.0329}$(18)
AutoEncoder	$0.8707_{\pm 0.0148}$(7)	$0.8186_{\pm 0.0140}$(3)	$0.4570_{\pm 0.0492}$(13)	$0.7890_{\pm 0.0327}$(13)
CBLOF	$0.8937_{\pm 0.0239}$(4)	$0.6153_{\pm 0.0182}$(13)	$0.4515_{\pm 0.0473}$(16)	$0.8082_{\pm 0.0357}$(7)
DIF	$0.6939_{\pm 0.0792}$(11)	$0.5566_{\pm 0.0306}$(17)	$0.4397_{\pm 0.0622}$(22)	$0.8060_{\pm 0.0362}$(8)
DiffInt	$0.9604_{\pm 0.0072}$(3)	$0.7041_{\pm 0.0072}$(11)	$0.5047_{\pm 0.0308}$(4)	$0.8193_{\pm 0.0185}$(5)
ECOD	$0.6357_{\pm 0.0142}$(15)	$0.7194_{\pm 0.0097}$(9)	$0.4511_{\pm 0.0486}$(17)	$0.8236_{\pm 0.0319}$(4)
GMM	$0.8067_{\pm 0.0087}$(10)	$0.7974_{\pm 0.0147}$(7)	$0.5013_{\pm 0.0483}$(5)	$0.7819_{\pm 0.0346}$(16)
HBOS	$0.8763_{\pm 0.0137}$(5)	$0.7924_{\pm 0.0236}$(8)	$0.4550_{\pm 0.0472}$(14)	$0.8287_{\pm 0.0309}$(**1**)
IForest	$0.8421_{\pm 0.0192}$(8)	$0.8394_{\pm 0.0139}$(**1**)	$0.4673_{\pm 0.0339}$(10)	$0.8247_{\pm 0.0345}$(2)
KDE	$0.9969_{\pm 0.0009}$(**1**)	$0.6064_{\pm 0.0140}$(15)	$0.6512_{\pm 0.0442}$(**1**)	$0.8150_{\pm 0.0298}$(6)
KPCA	$0.4975_{\pm 0.0319}$(17)	$0.5190_{\pm 0.0172}$(19)	$0.4890_{\pm 0.0492}$(8)	$0.4717_{\pm 0.0689}$(20)
LMDD	$0.4243_{\pm 0.0279}$(20)	$0.5985_{\pm 0.0418}$(16)	$0.4750_{\pm 0.0355}$(9)	$0.5193_{\pm 0.0305}$(19)
LOF	$0.4926_{\pm 0.0308}$(19)	$0.4242_{\pm 0.0150}$(21)	$0.4666_{\pm 0.0494}$(11)	$0.7974_{\pm 0.0365}$(10)
LUNAR	$0.9966_{\pm 0.0015}$(2)	$0.8229_{\pm 0.0234}$(2)	$0.5499_{\pm 0.0944}$(3)	$0.8239_{\pm 0.0310}$(3)
MCD	$0.6460_{\pm 0.0212}$(13)	$0.8001_{\pm 0.0260}$(6)	$0.4956_{\pm 0.0460}$(6)	$0.7812_{\pm 0.0400}$(17)
MO_GAAL	$0.3687_{\pm 0.1581}$(21)	$0.3253_{\pm 0.0590}$(22)	$0.4483_{\pm 0.0608}$(19)	$0.3027_{\pm 0.0591}$(22)
OCSVM	$0.6456_{\pm 0.0172}$(14)	$0.6178_{\pm 0.0142}$(12)	$0.4459_{\pm 0.0481}$(20)	$0.8005_{\pm 0.0279}$(9)
PCA	$0.5000_{\pm 0.0000}$(16)	$0.5500_{\pm 0.0125}$(18)	$0.4510_{\pm 0.0486}$(18)	$0.7858_{\pm 0.0339}$(15)
QMCD	$0.1303_{\pm 0.0179}$(23)	$0.7141_{\pm 0.0237}$(10)	$0.4217_{\pm 0.0446}$(23)	$0.3366_{\pm 0.0624}$(21)
SO_GAAL	$0.3658_{\pm 0.1716}$(22)	$0.3198_{\pm 0.0548}$(23)	$0.4457_{\pm 0.0656}$(21)	$0.3005_{\pm 0.0800}$(23)
Sampling	$0.8181_{\pm 0.0723}$(9)	$0.6115_{\pm 0.0168}$(14)	$0.4644_{\pm 0.0582}$(12)	$0.7964_{\pm 0.0368}$(11)
TCCM	$0.8725_{\pm 0.1032}$(6)	$0.8092_{\pm 0.0214}$(5)	$0.4899_{\pm 0.0480}$(7)	$0.7886_{\pm 0.0370}$(14)
VAE	$0.6583_{\pm 0.0277}$(12)	$0.8118_{\pm 0.0146}$(4)	$0.4516_{\pm 0.0489}$(15)	$0.7926_{\pm 0.0397}$(12)

TABLE XXV: AUPRC, small datasets (1/2).

Model	4_breastw	14_glass	15_Hepatitis	18_Ionosphere	21_Lymphography	29_Pima
ABOD	$0.2877_{\pm 0.0169}$(20)	$0.2252_{\pm 0.1340}$(8)	$0.3422_{\pm 0.1897}$(19)	$0.9189_{\pm 0.0188}$(9)	$0.6225_{\pm 0.2143}$(18)	$0.4735_{\pm 0.0308}$(16)
AutoEncoder	$0.9664_{\pm 0.0184}$(13)	$0.1502_{\pm 0.1025}$(16)	$0.6348_{\pm 0.1385}$(2)	$0.9416_{\pm 0.0131}$(5)	$0.8167_{\pm 0.1820}$(15)	$0.5387_{\pm 0.0181}$(7)
CBLOF	$0.9892_{\pm 0.0043}$(4)	$0.2286_{\pm 0.1182}$(7)	$0.5388_{\pm 0.1740}$(4)	$0.9201_{\pm 0.0374}$(8)	$0.9299_{\pm 0.1435}$(10)	$0.5244_{\pm 0.0279}$(10)
DIF	$0.6048_{\pm 0.0525}$(17)	$0.2519_{\pm 0.1434}$(4)	$0.4093_{\pm 0.1430}$(17)	$0.9367_{\pm 0.0108}$(6)	$0.4838_{\pm 0.2684}$(19)	$0.4144_{\pm 0.0207}$(20)
DiffInt	$0.9919_{\pm 0.0032}$(**1**)	$0.3294_{\pm 0.0800}$(**1**)	$0.5211_{\pm 0.0667}$(8)	$0.9574_{\pm 0.0073}$(2)	$0.9951_{\pm 0.0201}$(2)	$0.5668_{\pm 0.0149}$(3)
ECOD	$0.9869_{\pm 0.0053}$(7)	$0.1812_{\pm 0.1485}$(11)	$0.4933_{\pm 0.0920}$(10)	$0.6874_{\pm 0.0444}$(17)	$0.9867_{\pm 0.0292}$(4)	$0.4992_{\pm 0.0179}$(13)
GMM	$0.9693_{\pm 0.0139}$(12)	$0.2004_{\pm 0.1567}$(10)	$0.5098_{\pm 0.1610}$(9)	$0.9569_{\pm 0.0099}$(3)	$0.7933_{\pm 0.1614}$(16)	$0.5418_{\pm 0.0200}$(6)
HBOS	$0.9799_{\pm 0.0088}$(9)	$0.1794_{\pm 0.1047}$(12)	$0.5721_{\pm 0.0927}$(3)	$0.5333_{\pm 0.0867}$(19)	$0.9637_{\pm 0.0659}$(8)	$0.5991_{\pm 0.0253}$(2)
IForest	$0.9890_{\pm 0.0053}$(5)	$0.1543_{\pm 0.0638}$(14)	$0.5316_{\pm 0.1013}$(5)	$0.8529_{\pm 0.0338}$(13)	$0.9900_{\pm 0.0211}$(3)	$0.5572_{\pm 0.0257}$(4)
KDE	$0.9904_{\pm 0.0041}$(2)	$0.1668_{\pm 0.1048}$(13)	$0.4333_{\pm 0.0932}$(15)	$0.9323_{\pm 0.0105}$(7)	$0.9833_{\pm 0.0351}$(5)	$0.5295_{\pm 0.0295}$(9)
KPCA	$0.2457_{\pm 0.0273}$(21)	$0.0525_{\pm 0.0186}$(23)	$0.2266_{\pm 0.1709}$(21)	$0.3667_{\pm 0.0443}$(21)	$0.1209_{\pm 0.1796}$(20)	$0.3502_{\pm 0.0284}$(21)
LMDD	$0.7921_{\pm 0.1227}$(16)	$0.0951_{\pm 0.0897}$(22)	$0.2530_{\pm 0.1153}$(20)	$0.5672_{\pm 0.0604}$(18)	$0.7597_{\pm 0.1644}$(17)	$0.4895_{\pm 0.0585}$(14)
LOF	$0.3152_{\pm 0.0156}$(19)	$0.2320_{\pm 0.1363}$(6)	$0.4741_{\pm 0.1082}$(14)	$0.8282_{\pm 0.0611}$(14)	$0.9833_{\pm 0.0351}$(5)	$0.4409_{\pm 0.0227}$(18)
LUNAR	$0.9798_{\pm 0.0101}$(10)	$0.2542_{\pm 0.1299}$(2)	$0.4797_{\pm 0.1632}$(13)	$0.9677_{\pm 0.0119}$(**1**)	$0.8629_{\pm 0.1506}$(13)	$0.5502_{\pm 0.0212}$(5)
MCD	$0.9630_{\pm 0.0222}$(14)	$0.1475_{\pm 0.0490}$(17)	$0.4847_{\pm 0.1319}$(12)	$0.9534_{\pm 0.0115}$(4)	$0.8314_{\pm 0.1602}$(14)	$0.5054_{\pm 0.0295}$(11)
MO_GAAL	$0.2116_{\pm 0.0082}$(22)	$0.1409_{\pm 0.1880}$(19)	$0.2080_{\pm 0.1225}$(22)	$0.2876_{\pm 0.0459}$(22)	$0.0339_{\pm 0.0072}$(22)	$0.2784_{\pm 0.0480}$(22)
OCSVM	$0.9904_{\pm 0.0041}$(2)	$0.1249_{\pm 0.1108}$(21)	$0.5231_{\pm 0.1038}$(7)	$0.8001_{\pm 0.0329}$(15)	$0.9833_{\pm 0.0351}$(5)	$0.5028_{\pm 0.0288}$(12)
PCA	$0.9507_{\pm 0.0178}$(15)	$0.1454_{\pm 0.1055}$(18)	$0.4853_{\pm 0.1374}$(11)	$0.7236_{\pm 0.0393}$(16)	$0.9333_{\pm 0.1291}$(9)	$0.4678_{\pm 0.0199}$(17)
QMCD	$0.5168_{\pm 0.1374}$(18)	$0.2025_{\pm 0.0687}$(9)	$0.3999_{\pm 0.1074}$(18)	$0.3946_{\pm 0.0602}$(20)	$0.0378_{\pm 0.0084}$(21)	$0.6099_{\pm 0.0347}$(**1**)
SO_GAAL	$0.2116_{\pm 0.0082}$(22)	$0.1298_{\pm 0.1725}$(20)	$0.1719_{\pm 0.0960}$(23)	$0.2702_{\pm 0.0379}$(23)	$0.0338_{\pm 0.0071}$(23)	$0.2784_{\pm 0.0450}$(22)
Sampling	$0.9878_{\pm 0.0061}$(6)	$0.2452_{\pm 0.1173}$(5)	$0.4322_{\pm 0.1248}$(16)	$0.9093_{\pm 0.0388}$(10)	$0.8901_{\pm 0.1443}$(11)	$0.5312_{\pm 0.0416}$(8)
TCCM	$0.9718_{\pm 0.0148}$(11)	$0.2536_{\pm 0.1251}$(3)	$0.5267_{\pm 0.0975}$(6)	$0.9071_{\pm 0.0227}$(11)	$1.0000_{\pm 0.0000}$(**1**)	$0.4202_{\pm 0.0638}$(19)
VAE	$0.9806_{\pm 0.0108}$(8)	$0.1503_{\pm 0.0995}$(15)	$0.6372_{\pm 0.1153}$(**1**)	$0.8817_{\pm 0.0283}$(12)	$0.8796_{\pm 0.1430}$(12)	$0.4876_{\pm 0.0228}$(15)

TABLE XXVI: AUPRC, small datasets (2/2).

Model	37_Stamps	39_vertebral	42_WBC	43_WDBC	45_wine	46_WPBC
ABOD	$0.2120_{\pm 0.0573}$(19)	$0.1222_{\pm 0.0268}$(17)	$0.3421_{\pm 0.1683}$(18)	$0.2573_{\pm 0.1460}$(18)	$0.1717_{\pm 0.0927}$(20)	$0.2141_{\pm 0.0518}$(22)
AutoEncoder	$0.4350_{\pm 0.1051}$(13)	$0.1265_{\pm 0.0253}$(14)	$0.8106_{\pm 0.2272}$(13)	$0.7782_{\pm 0.2134}$(3)	$0.5529_{\pm 0.2822}$(6)	$0.2347_{\pm 0.0631}$(12)
CBLOF	$0.5528_{\pm 0.1300}$(2)	$0.1314_{\pm 0.0278}$(9)	$0.9480_{\pm 0.0504}$(5)	$0.6179_{\pm 0.2237}$(13)	$0.6318_{\pm 0.2372}$(2)	$0.2413_{\pm 0.0610}$(6)
DIF	$0.5181_{\pm 0.0937}$(4)	$0.1855_{\pm 0.0487}$(3)	$0.1239_{\pm 0.0324}$(19)	$0.0662_{\pm 0.0339}$(20)	$0.2498_{\pm 0.1465}$(18)	$0.2223_{\pm 0.0594}$(19)
DiffInt	$0.6095_{\pm 0.0647}$(1)	$0.1548_{\pm 0.0192}$(5)	$0.9658_{\pm 0.0466}$(1)	$0.5729_{\pm 0.1161}$(16)	$0.4625_{\pm 0.1262}$(11)	$0.2834_{\pm 0.0336}$(1)
ECOD	$0.4575_{\pm 0.1030}$(11)	$0.1310_{\pm 0.0239}$(10)	$0.9174_{\pm 0.0831}$(8)	$0.6875_{\pm 0.1531}$(6)	$0.3033_{\pm 0.0885}$(16)	$0.2158_{\pm 0.0474}$(21)
GMM	$0.4618_{\pm 0.0950}$(10)	$0.1291_{\pm 0.0219}$(11)	$0.6966_{\pm 0.1988}$(17)	$0.6851_{\pm 0.2442}$(8)	$0.6901_{\pm 0.2396}$(1)	$0.2392_{\pm 0.0704}$(8)
HBOS	$0.4570_{\pm 0.0663}$(12)	$0.1110_{\pm 0.0243}$(21)	$0.8305_{\pm 0.1530}$(12)	$0.7374_{\pm 0.1368}$(4)	$0.5273_{\pm 0.2059}$(9)	$0.2355_{\pm 0.0603}$(11)
IForest	$0.5178_{\pm 0.1048}$(5)	$0.1224_{\pm 0.0263}$(16)	$0.9201_{\pm 0.0784}$(7)	$0.6873_{\pm 0.1773}$(7)	$0.5717_{\pm 0.1853}$(5)	$0.2344_{\pm 0.0597}$(13)
KDE	$0.4759_{\pm 0.0940}$(8)	$0.1144_{\pm 0.0241}$(19)	$0.9502_{\pm 0.0568}$(3)	$0.6224_{\pm 0.1666}$(12)	$0.5357_{\pm 0.1809}$(7)	$0.2503_{\pm 0.0598}$(5)
KPCA	$0.0939_{\pm 0.0238}$(21)	$0.1707_{\pm 0.0392}$(4)	$0.1196_{\pm 0.1020}$(21)	$0.0208_{\pm 0.0054}$(21)	$0.1054_{\pm 0.0808}$(21)	$0.2364_{\pm 0.0843}$(10)
LMDD	$0.1085_{\pm 0.0399}$(20)	$0.1233_{\pm 0.0291}$(15)	$0.9302_{\pm 0.0867}$(6)	$0.7950_{\pm 0.1495}$(1)	$0.2445_{\pm 0.1397}$(19)	$0.2394_{\pm 0.0452}$(7)
LOF	$0.2330_{\pm 0.0598}$(18)	$0.1073_{\pm 0.0211}$(23)	$0.7069_{\pm 0.2313}$(16)	$0.6436_{\pm 0.1994}$(10)	$0.4505_{\pm 0.1887}$(12)	$0.2377_{\pm 0.0612}$(9)
LUNAR	$0.5381_{\pm 0.1321}$(3)	$0.1283_{\pm 0.0321}$(13)	$0.7695_{\pm 0.2384}$(15)	$0.7913_{\pm 0.1626}$(2)	$0.6064_{\pm 0.2519}$(3)	$0.2513_{\pm 0.0619}$(4)
MCD	$0.2714_{\pm 0.0547}$(17)	$0.1342_{\pm 0.0274}$(8)	$0.7814_{\pm 0.2436}$(14)	$0.4138_{\pm 0.1816}$(17)	$0.5772_{\pm 0.2406}$(4)	$0.2583_{\pm 0.0910}$(3)
MO_GAAL	$0.0723_{\pm 0.0240}$(22)	$0.2425_{\pm 0.0945}$(1)	$0.0282_{\pm 0.0047}$(22)	$0.0158_{\pm 0.0044}$(22)	$0.0440_{\pm 0.0147}$(22)	$0.2211_{\pm 0.0706}$(20)
OCSVM	$0.4249_{\pm 0.1001}$(14)	$0.1202_{\pm 0.0253}$(18)	$0.9502_{\pm 0.0568}$(3)	$0.6429_{\pm 0.1641}$(11)	$0.3484_{\pm 0.1777}$(14)	$0.2333_{\pm 0.0561}$(14)
PCA	$0.4083_{\pm 0.0799}$(15)	$0.1093_{\pm 0.0161}$(22)	$0.9009_{\pm 0.0873}$(9)	$0.6025_{\pm 0.1176}$(14)	$0.3114_{\pm 0.1194}$(15)	$0.2276_{\pm 0.0529}$(18)
QMCD	$0.4661_{\pm 0.0532}$(9)	$0.1144_{\pm 0.0252}$(19)	$0.1227_{\pm 0.0592}$(20)	$0.2214_{\pm 0.1521}$(19)	$0.2567_{\pm 0.1461}$(17)	$0.2330_{\pm 0.0558}$(15)
SO_GAAL	$0.0658_{\pm 0.0152}$(23)	$0.2141_{\pm 0.0595}$(2)	$0.0282_{\pm 0.0047}$(22)	$0.0158_{\pm 0.0044}$(22)	$0.0438_{\pm 0.0149}$(23)	$0.2100_{\pm 0.0529}$(23)
Sampling	$0.4803_{\pm 0.1574}$(7)	$0.1291_{\pm 0.0358}$(11)	$0.8885_{\pm 0.2170}$(10)	$0.5865_{\pm 0.1612}$(15)	$0.5280_{\pm 0.1743}$(8)	$0.2612_{\pm 0.0817}$(2)
TCCM	$0.3988_{\pm 0.0925}$(16)	$0.1363_{\pm 0.0268}$(7)	$0.9610_{\pm 0.0576}$(2)	$0.6813_{\pm 0.1950}$(9)	$0.5198_{\pm 0.2062}$(10)	$0.2295_{\pm 0.0547}$(16)
VAE	$0.4855_{\pm 0.0914}$(6)	$0.1434_{\pm 0.0283}$(6)	$0.8563_{\pm 0.1843}$(11)	$0.6896_{\pm 0.1968}$(5)	$0.4389_{\pm 0.1942}$(13)	$0.2283_{\pm 0.0544}$(17)

TABLE XXVII: AUPRC, medium datasets (1/3).

Model	2_annthyroid	6_cardio	7_Cardiotocography	12_fault	19_landsat	20_letter
ABOD	$0.1685_{\pm 0.0118}$(17)	$0.1865_{\pm 0.0217}$(19)	$0.2591_{\pm 0.0182}$(20)	$0.5010_{\pm 0.0160}$(10)	$0.2219_{\pm 0.0116}$(18)	$0.3120_{\pm 0.0422}$(6)
AutoEncoder	$0.5003_{\pm 0.0592}$(4)	$0.6195_{\pm 0.0650}$(12)	$0.4920_{\pm 0.0476}$(13)	$0.5937_{\pm 0.0420}$(5)	$0.2604_{\pm 0.0116}$(10)	$0.2659_{\pm 0.0448}$(7)
CBLOF	$0.1605_{\pm 0.0144}$(18)	$0.7159_{\pm 0.0405}$(7)	$0.5803_{\pm 0.0271}$(7)	$0.5318_{\pm 0.0543}$(9)	$0.3355_{\pm 0.0152}$(4)	$0.1951_{\pm 0.0339}$(9)
DIF	$0.4211_{\pm 0.0358}$(8)	$0.7767_{\pm 0.0339}$(4)	$0.5001_{\pm 0.0246}$(12)	$0.5894_{\pm 0.0342}$(6)	$0.2932_{\pm 0.0103}$(7)	$0.1555_{\pm 0.0476}$(10)
DiffInt	$0.3652_{\pm 0.0133}$(10)	$0.8114_{\pm 0.0213}$(2)	$0.6271_{\pm 0.0166}$(1)	$0.6639_{\pm 0.0128}$(2)	$0.3688_{\pm 0.0088}$(2)	$0.3954_{\pm 0.0306}$(2)
ECOD	$0.3479_{\pm 0.0222}$(11)	$0.6705_{\pm 0.0231}$(11)	$0.5820_{\pm 0.0238}$(6)	$0.3467_{\pm 0.0165}$(22)	$0.1769_{\pm 0.0043}$(21)	$0.0899_{\pm 0.0180}$(16)
GMM	$0.4365_{\pm 0.0291}$(7)	$0.6968_{\pm 0.0376}$(8)	$0.5269_{\pm 0.0250}$(9)	$0.5459_{\pm 0.0311}$(7)	$0.2060_{\pm 0.0064}$(20)	$0.3384_{\pm 0.0526}$(3)
HBOS	$0.3934_{\pm 0.0471}$(9)	$0.5475_{\pm 0.0316}$(14)	$0.4559_{\pm 0.0241}$(14)	$0.4442_{\pm 0.0669}$(15)	$0.3580_{\pm 0.0178}$(3)	$0.0867_{\pm 0.0190}$(18)
IForest	$0.4849_{\pm 0.0345}$(5)	$0.6925_{\pm 0.0521}$(9)	$0.5648_{\pm 0.0281}$(8)	$0.4958_{\pm 0.0382}$(11)	$0.2668_{\pm 0.0149}$(9)	$0.1011_{\pm 0.0206}$(15)
KDE	$0.1274_{\pm 0.0129}$(20)	$0.7267_{\pm 0.0282}$(6)	$0.6152_{\pm 0.0210}$(3)	$0.6196_{\pm 0.0279}$(4)	$0.2385_{\pm 0.0055}$(14)	$0.1506_{\pm 0.0325}$(11)
KPCA	$0.2036_{\pm 0.0217}$(13)	$0.0928_{\pm 0.0123}$(22)	$0.2136_{\pm 0.0213}$(21)	$0.3633_{\pm 0.0188}$(19)	$0.2150_{\pm 0.0096}$(19)	$0.0714_{\pm 0.0089}$(20)
LMDD	$0.1165_{\pm 0.0131}$(21)	$0.1569_{\pm 0.0189}$(20)	$0.4269_{\pm 0.0372}$(15)	$0.3829_{\pm 0.0243}$(18)	$0.2274_{\pm 0.0884}$(17)	$0.0638_{\pm 0.0079}$(21)
LOF	$0.1901_{\pm 0.0185}$(15)	$0.2026_{\pm 0.0235}$(18)	$0.2883_{\pm 0.0292}$(18)	$0.4002_{\pm 0.0206}$(17)	$0.2457_{\pm 0.0095}$(13)	$0.3357_{\pm 0.0444}$(4)
LUNAR	$0.4683_{\pm 0.0401}$(6)	$0.7791_{\pm 0.0356}$(3)	$0.6128_{\pm 0.0307}$(4)	$0.6845_{\pm 0.0358}$(1)	$0.5202_{\pm 0.0228}$(1)	$0.5013_{\pm 0.0652}$(1)
MCD	$0.5152_{\pm 0.0299}$(2)	$0.3805_{\pm 0.0375}$(15)	$0.3595_{\pm 0.0297}$(16)	$0.4347_{\pm 0.0468}$(16)	$0.2593_{\pm 0.0042}$(11)	$0.1998_{\pm 0.0415}$(8)
MO_GAAL	$0.1013_{\pm 0.0212}$(22)	$0.0918_{\pm 0.0954}$(23)	$0.2120_{\pm 0.0732}$(22)	$0.3535_{\pm 0.0403}$(20)	$0.3300_{\pm 0.0331}$(5)	$0.0577_{\pm 0.0097}$(22)
OCSVM	$0.1313_{\pm 0.0133}$(19)	$0.6923_{\pm 0.0300}$(10)	$0.6186_{\pm 0.0240}$(2)	$0.4703_{\pm 0.0216}$(13)	$0.1712_{\pm 0.0046}$(22)	$0.0807_{\pm 0.0169}$(19)
PCA	$0.2015_{\pm 0.0113}$(14)	$0.3667_{\pm 0.2827}$(16)	$0.2990_{\pm 0.1234}$(17)	$0.3415_{\pm 0.0173}$(23)	$0.1635_{\pm 0.0041}$(23)	$0.0878_{\pm 0.0205}$(17)
QMCD	$0.2606_{\pm 0.0275}$(12)	$0.3538_{\pm 0.0571}$(17)	$0.2713_{\pm 0.0218}$(19)	$0.4705_{\pm 0.0492}$(12)	$0.3152_{\pm 0.0113}$(6)	$0.1023_{\pm 0.0231}$(14)
SO_GAAL	$0.0981_{\pm 0.0196}$(23)	$0.0934_{\pm 0.0961}$(21)	$0.2083_{\pm 0.0727}$(23)	$0.3521_{\pm 0.0374}$(21)	$0.2349_{\pm 0.0285}$(16)	$0.0559_{\pm 0.0080}$(23)
Sampling	$0.1741_{\pm 0.0209}$(16)	$0.6043_{\pm 0.0843}$(13)	$0.5187_{\pm 0.0438}$(11)	$0.5323_{\pm 0.0531}$(8)	$0.2839_{\pm 0.0537}$(8)	$0.1489_{\pm 0.0309}$(12)
TCCM	$0.5279_{\pm 0.0935}$(1)	$0.8352_{\pm 0.0315}$(1)	$0.5254_{\pm 0.0509}$(10)	$0.6234_{\pm 0.0299}$(3)	$0.2533_{\pm 0.0126}$(12)	$0.3177_{\pm 0.0507}$(5)
VAE	$0.5096_{\pm 0.0311}$(3)	$0.7723_{\pm 0.0403}$(5)	$0.6065_{\pm 0.0273}$(5)	$0.4672_{\pm 0.0354}$(14)	$0.2372_{\pm 0.0057}$(15)	$0.1301_{\pm 0.0299}$(13)

TABLE XXVIII: AUPRC, medium datasets (2/3).

Model	27_PageBlocks	28_pendigits	30_satellite	31_satimage-2	38_thyroid	40_vowels
ABOD	$0.3540_{\pm 0.0282}$(18)	$0.0601_{\pm 0.0106}$(18)	$0.3867_{\pm 0.0150}$(20)	$0.1329_{\pm 0.0474}$(18)	$0.1251_{\pm 0.0139}$(21)	$0.4213_{\pm 0.1150}$(7)
AutoEncoder	$0.7179_{\pm 0.0330}$(4)	$0.4177_{\pm 0.0942}$(7)	$0.7908_{\pm 0.0112}$(6)	$0.8612_{\pm 0.0970}$(12)	$0.6788_{\pm 0.0687}$(5)	$0.5336_{\pm 0.0721}$(4)
CBLOF	$0.5343_{\pm 0.0463}$(12)	$0.4306_{\pm 0.1145}$(6)	$0.7996_{\pm 0.0198}$(5)	$0.9727_{\pm 0.0222}$(2)	$0.2328_{\pm 0.0654}$(16)	$0.3702_{\pm 0.1387}$(8)
DIF	$0.6854_{\pm 0.0315}$(6)	$0.5697_{\pm 0.1402}$(5)	$0.7803_{\pm 0.0134}$(7)	$0.9036_{\pm 0.0684}$(9)	$0.7491_{\pm 0.0391}$(**1**)	$0.2130_{\pm 0.0693}$(10)
DiffInt	$0.7673_{\pm 0.0121}$(**1**)	$0.8617_{\pm 0.0325}$(2)	$0.8168_{\pm 0.0060}$(2)	$0.9727_{\pm 0.0117}$(2)	$0.5704_{\pm 0.0412}$(10)	$0.7692_{\pm 0.0502}$(2)
ECOD	$0.5734_{\pm 0.0258}$(9)	$0.3003_{\pm 0.0431}$(11)	$0.5882_{\pm 0.0081}$(17)	$0.7695_{\pm 0.0614}$(15)	$0.6398_{\pm 0.0425}$(8)	$0.1090_{\pm 0.0263}$(16)
GMM	$0.7209_{\pm 0.0241}$(3)	$0.0879_{\pm 0.0160}$(16)	$0.7768_{\pm 0.0111}$(8)	$0.7337_{\pm 0.1016}$(16)	$0.6474_{\pm 0.0461}$(7)	$0.6293_{\pm 0.0983}$(3)
HBOS	$0.3019_{\pm 0.0160}$(19)	$0.2869_{\pm 0.0357}$(12)	$0.8109_{\pm 0.0092}$(4)	$0.8270_{\pm 0.0647}$(13)	$0.7392_{\pm 0.0772}$(3)	$0.1337_{\pm 0.0498}$(13)
IForest	$0.5398_{\pm 0.0328}$(11)	$0.3881_{\pm 0.0638}$(8)	$0.7662_{\pm 0.0140}$(10)	$0.9210_{\pm 0.0427}$(7)	$0.6635_{\pm 0.0942}$(6)	$0.1938_{\pm 0.0504}$(11)
KDE	$0.4864_{\pm 0.0316}$(15)	$0.7551_{\pm 0.0752}$(3)	$0.7479_{\pm 0.0115}$(12)	$0.9771_{\pm 0.0245}$(**1**)	$0.1606_{\pm 0.0532}$(20)	$0.1787_{\pm 0.0573}$(12)
KPCA	$0.1674_{\pm 0.0129}$(21)	$0.0228_{\pm 0.0061}$(21)	$0.3056_{\pm 0.0142}$(22)	$0.0628_{\pm 0.1632}$(20)	$0.0273_{\pm 0.0118}$(23)	$0.0378_{\pm 0.0092}$(21)
LMDD	$0.4825_{\pm 0.0221}$(16)	$0.0283_{\pm 0.0061}$(20)	$0.3157_{\pm 0.0782}$(21)	$0.0212_{\pm 0.0224}$(22)	$0.0696_{\pm 0.0161}$(22)	$0.0498_{\pm 0.0255}$(20)
LOF	$0.4083_{\pm 0.0256}$(17)	$0.0376_{\pm 0.0119}$(19)	$0.3932_{\pm 0.0145}$(19)	$0.0305_{\pm 0.0175}$(21)	$0.2889_{\pm 0.0935}$(13)	$0.4563_{\pm 0.0951}$(6)
LUNAR	$0.7430_{\pm 0.0304}$(2)	$0.9659_{\pm 0.0331}$(**1**)	$0.8405_{\pm 0.0110}$(**1**)	$0.9656_{\pm 0.0268}$(5)	$0.6089_{\pm 0.0850}$(9)	$0.8212_{\pm 0.0922}$(**1**)
MCD	$0.6209_{\pm 0.0226}$(8)	$0.0722_{\pm 0.0102}$(17)	$0.7754_{\pm 0.0136}$(9)	$0.7194_{\pm 0.0769}$(17)	$0.7096_{\pm 0.0493}$(4)	$0.0977_{\pm 0.0686}$(17)
MO_GAAL	$0.1085_{\pm 0.0568}$(23)	$0.0134_{\pm 0.0016}$(22)	$0.2918_{\pm 0.0194}$(23)	$0.0080_{\pm 0.0029}$(23)	$0.2146_{\pm 0.1110}$(17)	$0.0191_{\pm 0.0025}$(22)
OCSVM	$0.5083_{\pm 0.0317}$(14)	$0.3706_{\pm 0.0533}$(9)	$0.6837_{\pm 0.0107}$(15)	$0.9066_{\pm 0.0385}$(8)	$0.1701_{\pm 0.0539}$(19)	$0.0954_{\pm 0.0293}$(18)
PCA	$0.5327_{\pm 0.0308}$(13)	$0.2323_{\pm 0.0378}$(14)	$0.6038_{\pm 0.0099}$(16)	$0.8750_{\pm 0.0433}$(11)	$0.3753_{\pm 0.0389}$(11)	$0.0816_{\pm 0.0254}$(19)
QMCD	$0.2177_{\pm 0.0127}$(20)	$0.1684_{\pm 0.0255}$(15)	$0.8145_{\pm 0.0096}$(3)	$0.9346_{\pm 0.0343}$(6)	$0.3499_{\pm 0.0531}$(12)	$0.1307_{\pm 0.0329}$(14)
SO_GAAL	$0.1109_{\pm 0.0540}$(22)	$0.0134_{\pm 0.0013}$(22)	$0.4639_{\pm 0.1478}$(18)	$0.1315_{\pm 0.2360}$(19)	$0.2050_{\pm 0.0953}$(18)	$0.0191_{\pm 0.0027}$(22)
Sampling	$0.5436_{\pm 0.0792}$(10)	$0.3038_{\pm 0.2087}$(10)	$0.7445_{\pm 0.0386}$(13)	$0.8896_{\pm 0.1368}$(10)	$0.2644_{\pm 0.0744}$(15)	$0.3241_{\pm 0.0924}$(9)
TCCM	$0.7055_{\pm 0.0705}$(5)	$0.6903_{\pm 0.1344}$(4)	$0.7586_{\pm 0.0199}$(11)	$0.9695_{\pm 0.0281}$(4)	$0.2650_{\pm 0.0576}$(14)	$0.4667_{\pm 0.1105}$(5)
VAE	$0.6740_{\pm 0.0416}$(7)	$0.2768_{\pm 0.0448}$(13)	$0.7325_{\pm 0.0113}$(14)	$0.8211_{\pm 0.0622}$(14)	$0.7433_{\pm 0.0486}$(2)	$0.1170_{\pm 0.0567}$(15)

TABLE XXIX: AUPRC, medium datasets (3/3).

Model	41_Waveform	44_Wilt	47_yeast
ABOD	$0.0432_{\pm 0.0084}$(15)	$0.0548_{\pm 0.0024}$(9)	$0.3089_{\pm 0.0129}$(19)
AutoEncoder	$0.0787_{\pm 0.0299}$(7)	$0.0562_{\pm 0.0139}$(8)	$0.3197_{\pm 0.0182}$(13)
CBLOF	$0.2284_{\pm 0.0701}$(**1**)	$0.0387_{\pm 0.0030}$(19)	$0.3223_{\pm 0.0202}$(11)
DIF	$0.0873_{\pm 0.0246}$(6)	$0.0419_{\pm 0.0059}$(18)	$0.2961_{\pm 0.0133}$(23)
DiffInt	$0.0921_{\pm 0.0117}$(5)	$0.0798_{\pm 0.0060}$(5)	$0.3534_{\pm 0.0129}$(5)
ECOD	$0.0431_{\pm 0.0086}$(16)	$0.0428_{\pm 0.0048}$(15)	$0.3393_{\pm 0.0157}$(6)
GMM	$0.0426_{\pm 0.0086}$(18)	$0.0918_{\pm 0.0082}$(3)	$0.3096_{\pm 0.0171}$(16)
HBOS	$0.0508_{\pm 0.0073}$(13)	$0.0444_{\pm 0.0094}$(13)	$0.3334_{\pm 0.0165}$(8)
IForest	$0.0626_{\pm 0.0121}$(9)	$0.0453_{\pm 0.0033}$(12)	$0.3096_{\pm 0.0167}$(16)
KDE	$0.0696_{\pm 0.0097}$(8)	$0.0385_{\pm 0.0030}$(21)	$0.2974_{\pm 0.0143}$(21)
KPCA	$0.0320_{\pm 0.0048}$(21)	$0.0717_{\pm 0.0053}$(6)	$0.3807_{\pm 0.0174}$(**1**)
LMDD	$0.0323_{\pm 0.0044}$(20)	$0.0494_{\pm 0.0070}$(10)	$0.3337_{\pm 0.0185}$(7)
LOF	$0.1167_{\pm 0.0347}$(4)	$0.0426_{\pm 0.0029}$(16)	$0.3091_{\pm 0.0220}$(18)
LUNAR	$0.2161_{\pm 0.0711}$(2)	$0.0612_{\pm 0.0128}$(7)	$0.3226_{\pm 0.0206}$(10)
MCD	$0.0427_{\pm 0.0091}$(17)	$0.1547_{\pm 0.0106}$(**1**)	$0.3211_{\pm 0.0242}$(12)
MO_GAAL	$0.0212_{\pm 0.0064}$(23)	$0.0928_{\pm 0.0121}$(2)	$0.3647_{\pm 0.0406}$(3)
OCSVM	$0.0398_{\pm 0.0081}$(19)	$0.0387_{\pm 0.0032}$(19)	$0.3063_{\pm 0.0138}$(20)
PCA	$0.0466_{\pm 0.0082}$(14)	$0.0432_{\pm 0.0026}$(14)	$0.2967_{\pm 0.0169}$(22)
QMCD	$0.0594_{\pm 0.0072}$(11)	$0.0377_{\pm 0.0032}$(23)	$0.3283_{\pm 0.0192}$(9)
SO_GAAL	$0.0237_{\pm 0.0083}$(22)	$0.0902_{\pm 0.0115}$(4)	$0.3660_{\pm 0.0508}$(2)
Sampling	$0.1238_{\pm 0.0543}$(3)	$0.0420_{\pm 0.0030}$(17)	$0.3179_{\pm 0.0256}$(14)
TCCM	$0.0562_{\pm 0.0186}$(12)	$0.0385_{\pm 0.0049}$(21)	$0.3585_{\pm 0.0255}$(4)
VAE	$0.0616_{\pm 0.0121}$(10)	$0.0462_{\pm 0.0036}$(11)	$0.3138_{\pm 0.0179}$(15)

TABLE XXX: AUPRC, large datasets (1/2).

Model	1_ALOI	8_celeba	10_cover	11_donors	13_fraud	16_http
ABOD	$0.1185_{\pm 0.0111}$(2)	$0.0202_{\pm 0.0005}$(19)	$0.0288_{\pm 0.0013}$(17)	$0.0492_{\pm 0.0003}$(19)	$0.0159_{\pm 0.0011}$(18)	$0.0077_{\pm 0.0002}$(18)
AutoEncoder	$0.0419_{\pm 0.0033}$(8)	$0.0499_{\pm 0.0104}$(12)	$0.4044_{\pm 0.0611}$(3)	$0.3215_{\pm 0.0309}$(4)	$0.5606_{\pm 0.0488}$(**1**)	$0.7764_{\pm 0.1551}$(6)
CBLOF	$0.0406_{\pm 0.0034}$(9)	$0.0617_{\pm 0.0294}$(10)	$0.0772_{\pm 0.0259}$(10)	$0.3004_{\pm 0.0451}$(6)	$0.4031_{\pm 0.0301}$(8)	$0.4121_{\pm 0.0062}$(13)
DIF	$0.0438_{\pm 0.0026}$(5)	$0.0491_{\pm 0.0041}$(13)	$0.3492_{\pm 0.0373}$(4)	$0.2382_{\pm 0.0192}$(9)	$0.4106_{\pm 0.0472}$(7)	$0.3531_{\pm 0.0072}$(15)
DiffInt	$0.0807_{\pm 0.0033}$(4)	$0.0528_{\pm 0.0056}$(11)	$0.8938_{\pm 0.0073}$(**1**)	$0.9994_{\pm 0.0005}$(**1**)	$0.5503_{\pm 0.0193}$(2)	$0.8872_{\pm 0.1252}$(2)
ECOD	$0.0335_{\pm 0.0014}$(20)	$0.1007_{\pm 0.0055}$(4)	$0.1419_{\pm 0.0065}$(7)	$0.3098_{\pm 0.0022}$(5)	$0.2513_{\pm 0.0266}$(13)	$0.2311_{\pm 0.0028}$(16)
GMM	$0.0403_{\pm 0.0025}$(11)	$0.0910_{\pm 0.0042}$(6)	$0.1114_{\pm 0.0025}$(8)	$0.2779_{\pm 0.0016}$(8)	$0.4888_{\pm 0.0380}$(4)	$0.8511_{\pm 0.0052}$(4)
HBOS	$0.0343_{\pm 0.0013}$(17)	$0.1009_{\pm 0.0056}$(3)	$0.0352_{\pm 0.0019}$(16)	$0.1826_{\pm 0.0026}$(14)	$0.2132_{\pm 0.0351}$(14)	$0.5106_{\pm 0.1107}$(9)
IForest	$0.0337_{\pm 0.0016}$(19)	$0.0682_{\pm 0.0095}$(9)	$0.0384_{\pm 0.0049}$(14)	$0.2907_{\pm 0.0495}$(7)	$0.1265_{\pm 0.0279}$(17)	$0.3906_{\pm 0.0891}$(14)
KDE	$0.0423_{\pm 0.0034}$(6)	$0.0406_{\pm 0.0021}$(15)	$0.0495_{\pm 0.0016}$(12)	-	$0.3945_{\pm 0.0308}$(11)	-
KPCA	$0.0386_{\pm 0.0017}$(14)	-	-	-	-	-
LMDD	$0.0364_{\pm 0.0017}$(16)	-	-	-	-	-
LOF	$0.0901_{\pm 0.0074}$(3)	$0.0172_{\pm 0.0006}$(20)	$0.0136_{\pm 0.0012}$(19)	$0.1053_{\pm 0.0021}$(16)	$0.0018_{\pm 0.0001}$(19)	$0.0254_{\pm 0.0038}$(17)
LUNAR	$0.1491_{\pm 0.0110}$(**1**)	$0.0356_{\pm 0.0018}$(16)	$0.8678_{\pm 0.0130}$(2)	$0.9991_{\pm 0.0006}$(2)	$0.4674_{\pm 0.0555}$(5)	$0.5629_{\pm 0.3048}$(7)
MCD	$0.0341_{\pm 0.0017}$(18)	$0.0958_{\pm 0.0171}$(5)	$0.0158_{\pm 0.0004}$(18)	$0.1983_{\pm 0.1187}$(12)	$0.5063_{\pm 0.0249}$(3)	$0.8727_{\pm 0.0061}$(3)
MO_GAAL	$0.0290_{\pm 0.0016}$(23)	$0.0136_{\pm 0.0017}$(21)	$0.0101_{\pm 0.0008}$(20)	-	$0.0009_{\pm 0.0000}$(20)	-
OCSVM	$0.0421_{\pm 0.0034}$(7)	$0.0772_{\pm 0.0048}$(8)	$0.1093_{\pm 0.0046}$(9)	$0.2029_{\pm 0.0021}$(11)	$0.3949_{\pm 0.0306}$(10)	$0.4138_{\pm 0.0067}$(12)
PCA	$0.0380_{\pm 0.0018}$(15)	$0.1121_{\pm 0.0050}$(2)	$0.0732_{\pm 0.0015}$(11)	$0.1673_{\pm 0.0012}$(15)	$0.1506_{\pm 0.0159}$(16)	$0.4758_{\pm 0.0056}$(10)
QMCD	$0.0322_{\pm 0.0011}$(21)	$0.0222_{\pm 0.0006}$(17)	$0.0378_{\pm 0.0013}$(15)	$0.0937_{\pm 0.0010}$(17)	$0.2982_{\pm 0.0382}$(12)	$0.5394_{\pm 0.0175}$(8)
SO_GAAL	$0.0292_{\pm 0.0015}$(22)	$0.0214_{\pm 0.0010}$(18)	$0.0100_{\pm 0.0003}$(21)	$0.0513_{\pm 0.0082}$(18)	$0.0009_{\pm 0.0001}$(20)	$0.0038_{\pm 0.0001}$(19)
Sampling	$0.0400_{\pm 0.0036}$(12)	$0.0432_{\pm 0.0264}$(14)	$0.0465_{\pm 0.0212}$(13)	$0.2270_{\pm 0.0496}$(10)	$0.3992_{\pm 0.0286}$(9)	$0.4480_{\pm 0.1252}$(11)
TCCM	$0.0406_{\pm 0.0032}$(9)	$0.0833_{\pm 0.0067}$(7)	$0.2212_{\pm 0.1091}$(5)	$0.9181_{\pm 0.0839}$(3)	$0.4241_{\pm 0.0526}$(6)	$0.9790_{\pm 0.0097}$(**1**)
VAE	$0.0393_{\pm 0.0021}$(13)	$0.1142_{\pm 0.0058}$(**1**)	$0.1664_{\pm 0.0069}$(6)	$0.1888_{\pm 0.0413}$(13)	$0.1761_{\pm 0.0192}$(15)	$0.8511_{\pm 0.0060}$(4)

TABLE XXXI: AUPRC, large datasets (2/2).

Model	22_magic	23_mammography	32_shuttle	33_skin	34_smtp
ABOD	$0.7128_{\pm 0.0079}$(9)	$0.0240_{\pm 0.0016}$(21)	$0.1325_{\pm 0.0051}$(19)	$0.1771_{\pm 0.0013}$(21)	$0.0014_{\pm 0.0002}$(20)
AutoEncoder	$0.7687_{\pm 0.0135}$(3)	$0.3620_{\pm 0.0625}$(2)	$0.9496_{\pm 0.0117}$(12)	$0.4034_{\pm 0.1190}$(10)	$0.3247_{\pm 0.2130}$(10)
CBLOF	$0.7596_{\pm 0.0142}$(4)	$0.0785_{\pm 0.0120}$(19)	$0.9627_{\pm 0.0060}$(8)	$0.5527_{\pm 0.0108}$(4)	$0.4802_{\pm 0.1330}$(6)
DIF	$0.7236_{\pm 0.0113}$(7)	$0.2561_{\pm 0.0457}$(6)	$0.9701_{\pm 0.0083}$(5)	$0.4606_{\pm 0.0187}$(8)	$0.6147_{\pm 0.1393}$(3)
DiffInt	$0.8143_{\pm 0.0034}$(**1**)	$0.3010_{\pm 0.0216}$(4)	$0.9911_{\pm 0.0027}$(2)	$0.9647_{\pm 0.0053}$(**1**)	$0.7004_{\pm 0.0484}$(**1**)
ECOD	$0.5576_{\pm 0.0058}$(18)	$0.4516_{\pm 0.0447}$(**1**)	$0.9559_{\pm 0.0078}$(11)	$0.1988_{\pm 0.0010}$(18)	$0.5990_{\pm 0.1003}$(4)
GMM	$0.7397_{\pm 0.0068}$(5)	$0.2793_{\pm 0.0472}$(5)	$0.9404_{\pm 0.0117}$(13)	$0.4953_{\pm 0.0018}$(5)	$0.3155_{\pm 0.1184}$(11)
HBOS	$0.6460_{\pm 0.0072}$(14)	$0.2150_{\pm 0.0363}$(10)	$0.9668_{\pm 0.0108}$(6)	$0.3693_{\pm 0.0026}$(11)	$0.1901_{\pm 0.1312}$(12)
IForest	$0.6996_{\pm 0.0167}$(10)	$0.2438_{\pm 0.0504}$(9)	$0.9765_{\pm 0.0090}$(4)	$0.4900_{\pm 0.0084}$(6)	$0.0054_{\pm 0.0008}$(19)
KDE	$0.6595_{\pm 0.0057}$(11)	$0.1058_{\pm 0.0136}$(15)	$0.9596_{\pm 0.0069}$(10)	$0.2953_{\pm 0.0015}$(13)	$0.1038_{\pm 0.0424}$(15)
KPCA	$0.4320_{\pm 0.0090}$(20)	$0.0744_{\pm 0.0108}$(20)	$0.4836_{\pm 0.0176}$(18)	-	-
LMDD	$0.3822_{\pm 0.0185}$(21)	$0.0937_{\pm 0.0280}$(16)	$0.1166_{\pm 0.0400}$(20)	$0.1890_{\pm 0.0482}$(20)	$0.0717_{\pm 0.0759}$(16)
LOF	$0.5361_{\pm 0.0044}$(19)	$0.1153_{\pm 0.0104}$(13)	$0.0921_{\pm 0.0057}$(21)	$0.2298_{\pm 0.0015}$(17)	$0.1215_{\pm 0.0900}$(14)
LUNAR	$0.8041_{\pm 0.0064}$(2)	$0.3489_{\pm 0.0518}$(3)	$0.9939_{\pm 0.0074}$(**1**)	$0.9477_{\pm 0.0082}$(2)	$0.6708_{\pm 0.0870}$(2)
MCD	$0.6573_{\pm 0.0060}$(12)	$0.1425_{\pm 0.1684}$(12)	$0.8408_{\pm 0.0123}$(16)	$0.4762_{\pm 0.0021}$(7)	$0.0059_{\pm 0.0012}$(18)
MO_GAAL	$0.2282_{\pm 0.0055}$(23)	$0.0128_{\pm 0.0009}$(22)	$0.0371_{\pm 0.0007}$(22)	$0.1939_{\pm 0.0307}$(19)	$0.0003_{\pm 0.0001}$(21)
OCSVM	$0.6065_{\pm 0.0048}$(16)	$0.1124_{\pm 0.0148}$(14)	$0.9597_{\pm 0.0069}$(9)	$0.3467_{\pm 0.0021}$(12)	$0.1629_{\pm 0.0699}$(13)
PCA	$0.5939_{\pm 0.0065}$(17)	$0.1989_{\pm 0.0247}$(11)	$0.9164_{\pm 0.0063}$(15)	$0.1632_{\pm 0.0009}$(22)	$0.4109_{\pm 0.1316}$(9)
QMCD	$0.6329_{\pm 0.0082}$(15)	$0.0840_{\pm 0.0156}$(17)	$0.6512_{\pm 0.0358}$(17)	$0.2299_{\pm 0.0011}$(16)	$0.0222_{\pm 0.0313}$(17)
SO_GAAL	$0.2286_{\pm 0.0060}$(22)	$0.0128_{\pm 0.0009}$(22)	$0.0369_{\pm 0.0005}$(23)	$0.2632_{\pm 0.0935}$(14)	$0.0003_{\pm 0.0001}$(21)
Sampling	$0.7256_{\pm 0.0190}$(6)	$0.0840_{\pm 0.0224}$(17)	$0.9660_{\pm 0.0072}$(7)	$0.5764_{\pm 0.0818}$(3)	$0.4462_{\pm 0.1391}$(7)
TCCM	$0.7154_{\pm 0.0256}$(8)	$0.2440_{\pm 0.0541}$(8)	$0.9840_{\pm 0.0041}$(3)	$0.4232_{\pm 0.2331}$(9)	$0.5273_{\pm 0.1615}$(5)
VAE	$0.6508_{\pm 0.0137}$(13)	$0.2533_{\pm 0.0474}$(7)	$0.9336_{\pm 0.0114}$(14)	$0.2381_{\pm 0.0014}$(15)	$0.4244_{\pm 0.1402}$(8)

TABLE XXXII: AUPRC, high-dimensional datasets (1/2).

Model	3_backdoor	5_campaign	9_census	17_InternetAds	24_mnist	25_musk
ABOD	$0.0506_{\pm 0.0021}(17)$	$0.1690_{\pm 0.0041}(17)$	$0.0467_{\pm 0.0004}(10)$	$0.2707_{\pm 0.0321}(17)$	$0.2124_{\pm 0.0143}(15)$	$0.0349_{\pm 0.0173}(18)$
AutoEncoder	$0.8526_{\pm 0.0112}(4)$	$0.3426_{\pm 0.0116}(6)$	-	$0.8064_{\pm 0.0288}(2)$	$0.5906_{\pm 0.0384}(3)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
CBLOF	$0.0859_{\pm 0.0166}(14)$	$0.2479_{\pm 0.0200}(15)$	-	$0.5940_{\pm 0.0186}(7)$	$0.4009_{\pm 0.0413}(10)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
DIF	$0.7901_{\pm 0.0356}(7)$	$0.2486_{\pm 0.0192}(14)$	-	$0.2758_{\pm 0.0231}(16)$	$0.4513_{\pm 0.0344}(9)$	$0.9963_{\pm 0.0040}(11)$
DiffInt	$0.8897_{\pm 0.0092}(2)$	$0.3698_{\pm 0.0056}(2)$	$0.0767_{\pm 0.0005}(5)$	$0.5616_{\pm 0.0068}(10)$	$0.5209_{\pm 0.0264}(6)$	$0.9998_{\pm 0.0025}(10)$
ECOD	$0.1094_{\pm 0.0036}(13)$	$0.3696_{\pm 0.0115}(3)$	-	$0.5749_{\pm 0.0143}(9)$	$0.2070_{\pm 0.0118}(16)$	$0.9606_{\pm 0.0175}(14)$
GMM	$0.8557_{\pm 0.0138}(3)$	$0.3608_{\pm 0.0071}(4)$	-	$0.8295_{\pm 0.0217}(\mathbf{1})$	$0.5760_{\pm 0.0312}(4)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
HBOS	$0.0380_{\pm 0.0011}(19)$	$0.3864_{\pm 0.0133}(\mathbf{1})$	-	$0.2541_{\pm 0.0652}(18)$	$0.1543_{\pm 0.0124}(18)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
IForest	$0.0522_{\pm 0.0083}(15)$	$0.3123_{\pm 0.0210}(10)$	$0.0746_{\pm 0.0029}(6)$	$0.1604_{\pm 0.0224}(23)$	$0.3811_{\pm 0.0537}(11)$	$0.3308_{\pm 0.1853}(16)$
KDE	$0.7243_{\pm 0.0222}(8)$	$0.2834_{\pm 0.0111}(13)$	-	$0.7507_{\pm 0.0180}(5)$	$0.6664_{\pm 0.0270}(2)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
KPCA	-	$0.0878_{\pm 0.0021}(23)$	-	$0.2184_{\pm 0.0346}(20)$	$0.0939_{\pm 0.0064}(21)$	$0.0376_{\pm 0.0149}(17)$
LMDD	-	$0.1092_{\pm 0.0043}(20)$	-	$0.4432_{\pm 0.0183}(13)$	$0.0856_{\pm 0.0057}(23)$	$0.0299_{\pm 0.0031}(19)$
LOF	$0.2406_{\pm 0.0143}(9)$	$0.1358_{\pm 0.0030}(18)$	$0.0613_{\pm 0.0005}(8)$	$0.4235_{\pm 0.0355}(14)$	$0.2629_{\pm 0.0225}(14)$	$0.0259_{\pm 0.0051}(20)$
LUNAR	$0.8908_{\pm 0.0123}(\mathbf{1})$	$0.2889_{\pm 0.0122}(12)$	-	$0.8010_{\pm 0.0240}(4)$	$0.6665_{\pm 0.0287}(\mathbf{1})$	$1.0000_{\pm 0.0000}(\mathbf{1})$
MCD	$0.2331_{\pm 0.1673}(10)$	$0.3155_{\pm 0.0121}(9)$	$0.1879_{\pm 0.0548}(\mathbf{1})$	$0.2296_{\pm 0.0375}(19)$	$0.3740_{\pm 0.1058}(12)$	$0.9951_{\pm 0.0116}(12)$
MO_GAAL	$0.0519_{\pm 0.0835}(16)$	$0.0998_{\pm 0.0183}(22)$	-	$0.1865_{\pm 0.0175}(22)$	$0.1068_{\pm 0.0420}(20)$	$0.0199_{\pm 0.0031}(22)$
OCSVM	$0.1114_{\pm 0.0038}(12)$	$0.3116_{\pm 0.0120}(11)$	-	$0.5813_{\pm 0.0144}(8)$	$0.4554_{\pm 0.0251}(8)$	$0.8724_{\pm 0.0631}(15)$
PCA	$0.0246_{\pm 0.0007}(20)$	$0.1134_{\pm 0.0017}(19)$	$0.0617_{\pm 0.0005}(7)$	$0.2987_{\pm 0.0112}(15)$	$0.0922_{\pm 0.0043}(22)$	$0.9999_{\pm 0.0004}(9)$
QMCD	$0.0128_{\pm 0.0004}(21)$	$0.3406_{\pm 0.0152}(7)$	$0.1194_{\pm 0.0013}(2)$	$0.7418_{\pm 0.0328}(6)$	$0.1640_{\pm 0.0134}(17)$	$0.0159_{\pm 0.0016}(23)$
SO_GAAL	$0.0443_{\pm 0.0635}(18)$	$0.1026_{\pm 0.0209}(21)$	$0.0600_{\pm 0.0038}(9)$	$0.1898_{\pm 0.0100}(21)$	$0.1099_{\pm 0.0404}(19)$	$0.0203_{\pm 0.0033}(21)$
Sampling	$0.1406_{\pm 0.1835}(11)$	$0.2230_{\pm 0.0340}(16)$	-	$0.5368_{\pm 0.0269}(12)$	$0.3388_{\pm 0.1105}(13)$	$0.9913_{\pm 0.0105}(13)$
TCCM	$0.8419_{\pm 0.0155}(6)$	$0.3291_{\pm 0.0169}(8)$	$0.0950_{\pm 0.0033}(4)$	$0.5548_{\pm 0.0152}(11)$	$0.5063_{\pm 0.0472}(7)$	$1.0000_{\pm 0.0000}(\mathbf{1})$
VAE	$0.8466_{\pm 0.0085}(5)$	$0.3510_{\pm 0.0091}(5)$	$0.1119_{\pm 0.0020}(3)$	$0.8043_{\pm 0.0291}(3)$	$0.5727_{\pm 0.0339}(5)$	$1.0000_{\pm 0.0000}(\mathbf{1})$

TABLE XXXIII: AUPRC, high-dimensional datasets (2/2).

Model	26_optdigits	35_SpamBase	36_speech	arrhythmia
ABOD	$0.0562_{\pm 0.0392}(12)$	$0.3895_{\pm 0.0078}(20)$	$0.0247_{\pm 0.0103}(8)$	$0.3791_{\pm 0.0363}(18)$
AutoEncoder	$0.1018_{\pm 0.0108}(7)$	$0.7571_{\pm 0.0189}(2)$	$0.0220_{\pm 0.0086}(13)$	$0.4081_{\pm 0.0651}(15)$
CBLOF	$0.1253_{\pm 0.0305}(6)$	$0.4588_{\pm 0.0198}(12)$	$0.0217_{\pm 0.0084}(15)$	$0.4896_{\pm 0.0815}(8)$
DIF	$0.0567_{\pm 0.0228}(11)$	$0.4532_{\pm 0.0279}(14)$	$0.0161_{\pm 0.0048}(20)$	$0.4995_{\pm 0.0716}(7)$
DiffInt	$0.3739_{\pm 0.0616}(3)$	$0.5853_{\pm 0.0083}(11)$	$0.0233_{\pm 0.0046}(10)$	$0.5067_{\pm 0.0406}(5)$
ECOD	$0.0385_{\pm 0.0051}(17)$	$0.6205_{\pm 0.0105}(10)$	$0.0227_{\pm 0.0094}(11)$	$0.5172_{\pm 0.0701}(3)$
GMM	$0.0719_{\pm 0.0093}(10)$	$0.7368_{\pm 0.0186}(5)$	$0.0329_{\pm 0.0182}(2)$	$0.3955_{\pm 0.0553}(17)$
HBOS	$0.1829_{\pm 0.0261}(5)$	$0.7049_{\pm 0.0365}(8)$	$0.0250_{\pm 0.0114}(5)$	$0.5092_{\pm 0.0746}(4)$
IForest	$0.0992_{\pm 0.0195}(9)$	$0.7811_{\pm 0.0239}(\mathbf{1})$	$0.0244_{\pm 0.0150}(9)$	$0.5250_{\pm 0.0838}(2)$
KDE	$0.8656_{\pm 0.0478}(\mathbf{1})$	$0.4445_{\pm 0.0127}(17)$	$0.0465_{\pm 0.0207}(\mathbf{1})$	$0.5008_{\pm 0.0690}(6)$
KPCA	$0.0317_{\pm 0.0060}(18)$	$0.4376_{\pm 0.0188}(18)$	$0.0197_{\pm 0.0073}(19)$	$0.1603_{\pm 0.0393}(21)$
LMDD	$0.0273_{\pm 0.0036}(20)$	$0.4499_{\pm 0.0337}(16)$	$0.0199_{\pm 0.0056}(18)$	$0.2404_{\pm 0.0510}(19)$
LOF	$0.0524_{\pm 0.0200}(13)$	$0.3528_{\pm 0.0058}(21)$	$0.0249_{\pm 0.0111}(6)$	$0.4532_{\pm 0.0668}(12)$
LUNAR	$0.8328_{\pm 0.0875}(2)$	$0.7526_{\pm 0.0346}(3)$	$0.0311_{\pm 0.0148}(3)$	$0.5266_{\pm 0.0686}(\mathbf{1})$
MCD	$0.0388_{\pm 0.0063}(16)$	$0.7187_{\pm 0.0498}(7)$	$0.0248_{\pm 0.0108}(7)$	$0.4173_{\pm 0.0752}(14)$
MO_GAAL	$0.0237_{\pm 0.0072}(22)$	$0.3057_{\pm 0.0269}(22)$	$0.0143_{\pm 0.0030}(22)$	$0.1156_{\pm 0.0318}(22)$
OCSVM	$0.0401_{\pm 0.0056}(15)$	$0.4521_{\pm 0.0128}(15)$	$0.0218_{\pm 0.0088}(14)$	$0.4846_{\pm 0.0678}(10)$
PCA	$0.0298_{\pm 0.0043}(19)$	$0.4149_{\pm 0.0104}(19)$	$0.0216_{\pm 0.0085}(16)$	$0.4433_{\pm 0.0536}(13)$
QMCD	$0.0163_{\pm 0.0024}(23)$	$0.6210_{\pm 0.0247}(9)$	$0.0133_{\pm 0.0024}(23)$	$0.2007_{\pm 0.0700}(20)$
SO_GAAL	$0.0240_{\pm 0.0080}(21)$	$0.3042_{\pm 0.0237}(23)$	$0.0144_{\pm 0.0036}(21)$	$0.1119_{\pm 0.0279}(23)$
Sampling	$0.0996_{\pm 0.0674}(8)$	$0.4545_{\pm 0.0162}(13)$	$0.0225_{\pm 0.0109}(12)$	$0.4849_{\pm 0.0726}(9)$
TCCM	$0.2340_{\pm 0.1204}(4)$	$0.7235_{\pm 0.0224}(6)$	$0.0284_{\pm 0.0127}(4)$	$0.4658_{\pm 0.0748}(11)$
VAE	$0.0403_{\pm 0.0077}(14)$	$0.7460_{\pm 0.0198}(4)$	$0.0216_{\pm 0.0085}(16)$	$0.4004_{\pm 0.0621}(16)$